Magnetic damping of standing waves in rectangular geometries

Pranav Hegde,[1] Wietze Herreman,[2] Thomas Gundrum,[1] and Gerrit Maik Horstmann[1,*]

[1] *Helmholtz-Zentrum Dresden-Rossendorf, Bautzner Landstr. 400, 01328 Dresden, Germany*
[2] *Université Paris Saclay, CNRS, FAST, Orsay, F-91405, France*

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We present a theoretical formulation of the magnetic damping in a free surface system in the presence of a vertical magnetic field. Motion of the conducting medium coupled with the perpendicular magnetic field leads to the growth of induced eddy currents, which generates heat in the form of Ohmic dissipation. This results in the dissipation of the mechanical energy of an initially excited linear sloshing wave. Cases with perfectly conducting walls have a relatively simple distribution of eddy currents and have a straightforward solution towards the decay rate of waves in the presence of a magnetic field. However, this assumption is not suited for practical applications with non-conducting walls or for a mixed configuration with conducting and non-conducting walls, where the current distribution is confined within the walls leading to a significantly lower magnetic damping value. In the present work, we extend the magnetic damping theory to account for free-surface waves with arbitrary wave modes at non-shallow depths enclosed in a domain with electrically conducting, insulating and mixed boundary conditions. Moreover, we give an analytical description of the Hartmann and Shercliff boundary layers and their influence on dissipation at the different boundaries. We validate the analytical results using existing analytical and experimental studies including our own setup for non-shallow free surface waves.

I. INTRODUCTION

In recent years, we have seen the emergence of techniques which use electromagnetism for the processing conductive fluids, such as liquid metals, molten salts or semiconductor materials. Electromagnetic processing of materials (EPM) has been extensively used in various metallurgical devices for heating and stirring liquid metals [1, 2]. The electromagnetic fields produced by an alternating current (AC), are used to induce electric currents in the medium. This process, known as induction heating, uses the inherent resistance of the material to melt it. Moreover, the current also interacts with the recently formed fluid medium, producing Lorentz forces; the oscillating part of which are known to generate free surface/interfacial waves. In continuous casting of steel, electromagnetic stirrers make use of a rotating magnetic field to induce a swirling motion in the molten metal to homogenize the temperature distribution, and thereby also the microstructure of the cast [3]. In the context of induction heaters, the resultant velocity fluctuations reduce the overall quality of the molten product, mixing debris or atmospheric gases into the liquid. Whereas, in electromagnetic stirrers, the secondary vertical motion (sloshing) at certain resonant AC frequencies causes slag entrainment and surface cracking. Therefore, in these industrial processes, it is necessary to suppress the undesirable effects produced by an AC magnetic field.

It is well established that the magnetic field produced by direct current (DC) has a damping effect on the fluid flow [4–6]. However, this effect is not isotropic and largely depends on the orientation of the magnetic field with respect to the local velocity vector of the fluid. For example, a vertical magnetic field $\boldsymbol{B}_0$ imposed on a conducting liquid in motion results in the generation of an induced current density

$$\boldsymbol{j} = \sigma(-\boldsymbol{\nabla}\varphi + \boldsymbol{U} \times \boldsymbol{B}_0), \tag{1}$$

which is orthogonal to the average velocity $\boldsymbol{U}$ and the magnetic field. The term σ is the electrical conductivity and φ, the electric potential of the fluid. The internal resistance of the fluid contributes to the Ohmic losses in the system, which decrease the overall kinetic energy of the fluid. According to the inviscid equation of motion [6, chapter 5],

$$\frac{\mathrm{d}}{\mathrm{d}t}\int_V \frac{1}{2}\rho \boldsymbol{U}^2 \mathrm{d}V = -\frac{1}{\sigma}\int_V \boldsymbol{j}^2 \mathrm{d}V \tag{2}$$

where ρ is the density of the fluid occupying a volume V. The induced currents also interact with the magnetic field producing Lorentz forces: $\mathrm{f_L} = \boldsymbol{j} \times \boldsymbol{B}_0$. At first sight, it appears from this equation that motion normal to the field lines experiences braking due to Lorentz forces whereas motion parallel to the field lines is unopposed due to

* g.horstmann@hzdr.de

$\boldsymbol{U} \times \boldsymbol{B}_0 = 0$. This is, however, a gross oversimplification of the phenomenon as the effect of the electric potential is ignored. To conserve the overall charge in the system, the current loops must close either within the fluid or in the walls depending on the electrical boundary conditions. To do so, the current must flow back through other parts of the system, and the redistributed current generates Lorentz forces that oppose the flow even in directions parallel to the field lines. This must be considered when theoretically modeling magnetohydrodynamic (MHD) effects in liquid metals.

One of the early theoretical studies on magnetic field induced damping was by Moffatt [7]. The study focused on the suppression of isotropic turbulence in a conducting fluid in the presence of a uniformly applied magnetic field and highlighted the contribution of Ohmic dissipation in damping. Shercliff [8] showed that the vertical magnetic field acts isotropically, whereas a horizontal magnetic field produces an anisotropic effect on the free surface. The action of the Lorentz force depends on the orientation of the wave relative to the field. The damping of gravity waves due to Ohmic dissipation were also explored by Rivat *et al.* [9] with respect to a funnel used in the metal casting process. While the damping nature of a static field is already known, magnetic damping can also appear as a second-order effect that can suppress wave instabilities generated by an AC magnetic field, as shown by Fautrelle and Sneyd [10]. There exists a strong balance between magnetic forcing and damping when the field strength is high enough to trigger the parametric instability in a cylindrical geometry. The authors also gave a rough estimate regarding the formation of the Hartmann layer, which occurs at the boundary perpendicular to the magnetic field.

There have been many experiments that have investigated the action of AC and DC magnetic fields on liquid metal free surface waves. Fautrelle *et al.* [11] focused on the stabilizing effect of a non-vertical DC magnetic field. The authors applied a horizontal transverse DC field to a mercury bath in a rectangular steel container. It was observed that the application of the static field suppressed all the free surface fluctuations and instead gave rise to a single traveling wave. Bojarevičs *et al.* [12] investigated the influence of a static magnetic field in the vertical, transverse and coplanar orientations on inductively heated melts. The vertical orientation of the field proved to be the most efficient in damping of the flow velocities and surface deformations, whereas the coplanar orientation showed no overall stabilization. An increase in velocity fluctuations in the melt was observed instead. Fisher *et al.* [13] showed that transverse magnetic fields perpendicular to wave propagation in shallow cells have no effects on wave damping, with or without the presence of external currents. In addition to the magnetic field orientation, the type of electrical boundary condition at the surrounding walls is an important factor to be considered. This was addressed by Sreenivasan *et al.* [14], who conducted damping experiments in conducting and insulating walls in the presence of a vertical magnetic field. The presence of non-conducting sidewalls was found to reduce the rate of magnetic damping. The stabilizing effect of magnetic damping is of significant importance in the design of laboratory-scale wave experiments, which often make use of high-strength magnetic fields [15–18]. These are relevant for cases where the objective is to excite or destabilize a wave with the use of external currents. To do so, the power injected into the system, whether in the form of input current or external magnetic field should exceed the total dissipation, either viscous or magnetic.

Returning to the aim of the present work: we present a complete theoretical model of magnetic damping in a fluid undergoing linear wave motion in a rectangular domain. Magnetic damping in cylindrical geometries has already been well studied [19, 20]. The theoretical model put forth by Sreenivasan *et al.* [14] takes into account only simple, unidirectional waves in rectangular geometries with shallow depths. Building on this work, we have expanded their description to derive analytical representations of induced eddy current distributions and magnetic damping rates for arbitrary wave modes in arbitrary rectangular geometries with homogeneous as well as mixed electrical boundary conditions. We also consider the effect of the Hartmann and Shercliff layers formed at the boundaries. A particular case of our analytical solution for the Ohmic dissipation rate has also been tested experimentally using two different electrical boundary conditions. This study is also significant in finalizing the damping model discussed in Hegde *et al.* [21], which addresses long-wave interfacial instabilities that limit the operation in aluminum reduction cells. Magnetic damping in addition to viscous damping is largely unexplored regarding reduction cells.

The manuscript is organised as follows: In Sec. II we present the theoretical formulation, starting with the MHD equations and all the different cases based on the electrical boundary conditions (Sec. II A), and highlight the solution for the induced electric potentials (Sec. II B). In Sec. II C, we provide a quantitative description of the eddy current distribution in the fluid medium, and in Sec. II D, the description for the bulk dissipation due to the vertical magnetic field. In Sec. II E, we discuss boundary layer effects and the transition of the viscous sub-layer to the Hartmann and Shercliff layers in the presence of a magnetic field. We consolidate all the theoretical results in Sec. III and compare it with the theoretical and experimental results of Sreenivasan *et al.* [14] in Sec. III A. In Sec. IV, we give a description of the experimental model and the magnetic damping measurements. We conclude our study and discuss open questions in Sec. V.

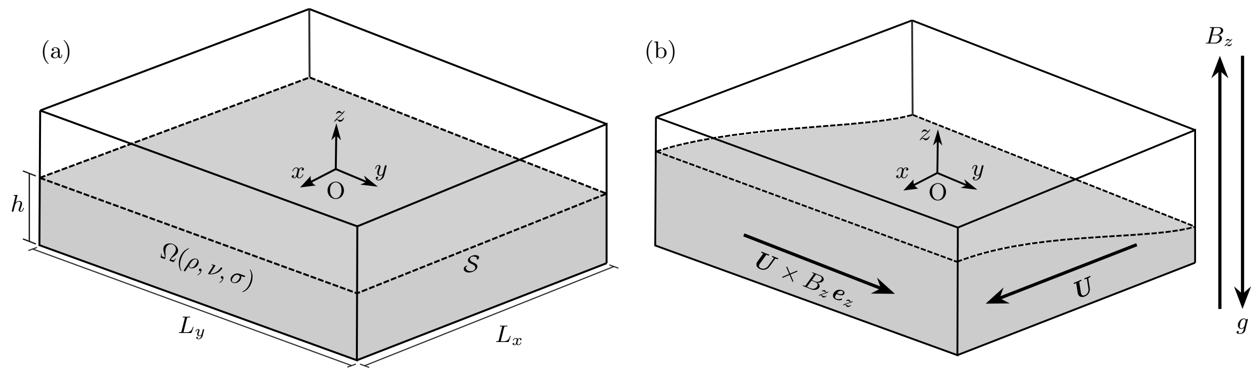


FIG. 1. The three-dimensional sketch describing the a) single-layer system Ω of depth h, with density, kinematic viscosity and electrical conductivity denoted as ρ, ν and σ respectively, and b) the mechanism of induction ($\propto \boldsymbol{U} \times B_z\boldsymbol{e}_z$) due to the motion of the free surface $\mathcal{S}$ in the presence of a vertical magnetic field, B_z. The acceleration due to gravity g acts in the downward direction along z-axis.

II. THEORETICAL FORMULATION

A. The governing equations

We consider a single-layer system consisting of a conducting liquid Ω where the free surface is permitted to move along the vertical axis. The fluid, which has density ρ, kinematic viscosity ν, electrical conductivity σ, is bound by a rectangular domain of length L_x and width L_y, with a non-shallow depth of h, as shown in Fig. 1(a). The free surface $\mathcal{S}$ is threaded by a constant, vertical magnetic field along the positive z-axis. The average fluid flow $\boldsymbol{U}$ due to wave motion coupled with the vertical component of the magnetic field, $B_z\boldsymbol{e}_z$ produces induced currents proportional to $\boldsymbol{U} \times B_z\boldsymbol{e}_z$, a key contributor to Ohmic dissipation. This is illustrated in Fig. 1(b). The induced magnetic field due to induced currents are negligible due to higher order effects and as a result, are ignored in this study.

We denote the following quantities $\boldsymbol{u}$, p, $\boldsymbol{J}$, and Ψ for the local velocity, pressure, induced current density and the electric potential due to induction, and write the governing equations as follows:

$$\rho\partial_t\boldsymbol{u} + \boldsymbol{\nabla}p = \boldsymbol{J} \times B_z\boldsymbol{e}_z + \rho\nu\nabla^2\boldsymbol{u}, \tag{3a}$$

$$\boldsymbol{\nabla}\cdot\boldsymbol{u} = 0, \tag{3b}$$

$$\boldsymbol{J} = \sigma\big(-\boldsymbol{\nabla}\Psi + \boldsymbol{u} \times B_z\boldsymbol{e}_z\big), \tag{3c}$$

$$\boldsymbol{\nabla}\cdot\boldsymbol{J} = 0. \tag{3d}$$

The Eqs. (3a–3d) are the linearized momentum, continuity, generalized Ohm's law and continuity of charge equation in order. The source term $\boldsymbol{J} \times B_z\boldsymbol{e}_z$ is the Lorentz force due to the coupling of induced current density. We assume the impermeability condition at the wall i.e. velocity perpendicular ($\boldsymbol{u}\cdot\boldsymbol{n} = 0$) to the wall is zero. The vertical displacement of the free surface at an instantaneous time t is

$$\eta(x,y,t) = \mathrm{i}A_{mn}\cos\left[\frac{m\pi}{L_x}\left(x + \frac{L_x}{2}\right)\right]\cos\left[\frac{n\pi}{L_y}\left(y + \frac{L_y}{2}\right)\right]\mathrm{e}^{\mathrm{i}\omega t}, \tag{4}$$

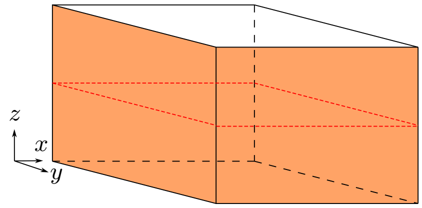
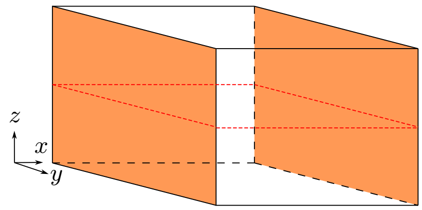
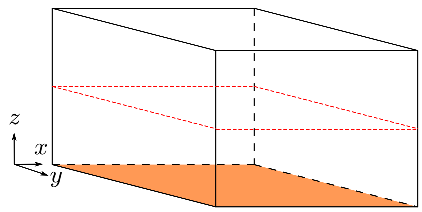
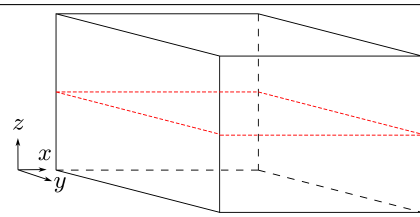
where, A_{mn} is the initial amplitude and $m, n \in \mathbb{N}_0 \neq (0,0)$ refer to the mode indices counting half wavelengths of the plane wave in the x and y direction respectively. In eq. (4), ω refers to the angular frequency of the linear wave oscillation

$$\omega = \sqrt{gk\tanh\left[kh\right]}\,, \tag{5}$$

Here, $k = \pi\sqrt{\left(m/L_x\right)^2 + \left(n/L_y\right)^2}$ is the angular wave number and g, the acceleration due to gravity. In a classical gravity wave system, we assume the fluid to be both incompressible and irrotational, with velocity given by $\boldsymbol{u} = \boldsymbol{\nabla}\phi$, where ϕ is the scalar velocity potential of the fluid. Solving the Laplace equation $\boldsymbol{\nabla}^2\phi = 0$ gives the following solution:

$$\phi = -A_{mn}\frac{\omega}{k}\frac{\cosh\left[k\big(z+h\big)\right]}{\sinh\left[kh\right]}\cos\left[\frac{m\pi}{L_x}\left(x + \frac{L_x}{2}\right)\right]\cos\left[\frac{n\pi}{L_y}\left(y + \frac{L_y}{2}\right)\right]\mathrm{e}^{\mathrm{i}\omega t}. \tag{6}$$

TABLE I. The four different cases adopted in the current study and its respective boundary conditions. The conducting walls are highlighted in orange while the insulating walls are in white. The free surface is marked in red.

Case No.	Case name	Boundary conditions	
1	B.C 1		$\Psi\big\|_{x=\pm 0.5L_x} = \Psi\big\|_{y=\pm 0.5L_y} = 0,$ $\Psi\|_{z=-h} = 0,\ \partial_z \Psi\|_{z=0} = 0.$
2	B.C 2		$\Psi\big\|_{x=\pm 0.5L_x} = 0,$ $\partial_y \Psi\big\|_{y=\pm 0.5L_y} = -\partial_x \phi B_z,$ $\partial_z \Psi\|_{z=0} = 0,\ \partial_z \Psi\|_{z=-h} = 0.$
3	B.C 3		$\partial_x \Psi\big\|_{x=\pm 0.5L_x} = \partial_y \phi B_z,$ $\partial_y \Psi\big\|_{y=\pm 0.5L_y} = -\partial_x \phi B_z,$ $\partial_z \Psi\|_{z=0} = 0,\ \Psi\|_{z=-h} = 0.$
4	B.C 4		$\partial_x \Psi\big\|_{x=\pm 0.5L_x} = \partial_y \phi B_z,$ $\partial_y \Psi\big\|_{y=\pm 0.5L_y} = -\partial_x \phi B_z,$ $\partial_z \Psi\|_{z=0} = 0,\ \partial_z \Psi\|_{z=-h} = 0.$

Certain assumptions must be made before conducting the theoretical analysis. Firstly, the effects of surface tension are ignored. Secondly, that only the vertical component of the DC magnetic field is of interest in the current study. Thirdly, we undertake the low magnetic Reynolds number approximation:

$$\mathrm{R_m} = \mu_0 \sigma \omega L^2 \ll 1, \tag{7}$$

which states that the induced magnetic fields due to the induced currents are negligible compared to the magnitude of the external magnetic field B_z. In the above equation, μ_0 is the permeability of free space and L is the characteristic length. Note that the wave velocity is expressed as the product of wave frequency ω and L. The effect of B_z can also be characterized by the Hartmann number,

$$\mathrm{Ha} = B_z L \sqrt{\frac{\sigma}{\rho \nu}}, \tag{8}$$

which is the ratio of electromagnetic to viscous forces in a domain of characteristic length L. The Hartmann number plays a crucial role in the transition of the viscous boundary into the Hartmann layer, where the circulating electric currents cut the magnetic fields producing an electromotive force suppressing fluid motion which results in a "braking" effect. This is addressed in Sec. II E.

B. The electric potential

To determine the Ohmic dissipation, it is necessary to first calculate the induced current density $\boldsymbol{J}$ in the bulk fluid flow, which has known quantities $\boldsymbol{u} = \boldsymbol{\nabla}\phi$, and unknown quantities like the induced electric potential Ψ. To this end, we substitute the generalized Ohm's law [Eq. (3c)] in the conservation of induced current density [Eq. (3d)]:

$$\boldsymbol{\nabla}\cdot\boldsymbol{J}=0\Longrightarrow-\nabla^2\Psi+\boldsymbol{\nabla}\cdot\big(\boldsymbol{u}\times B_z\boldsymbol{e}_z\big)=0, \tag{9}$$

to obtain the Laplace equation

$$\nabla^2\Psi=0, \tag{10}$$

the solution of which, largely depends on the electrical boundary conditions, which vary depending on the model of the industrial application. To address this. we introduce four distinct cases namely B.C 1, B.C 2, B.C 3 and B.C 4 as shown in Table I. In B.C 1, all bounding walls are perfect electrical conductors which act as an ideal electrode ($\Psi|_{\text{wall}}=0$). In B.C 2, only a pair of sidewalls perpendicular to the x-axis are perfect conductors. The rest of the walls are perfect insulators, which prevents the flow of current into the wall ($\boldsymbol{J}\cdot\boldsymbol{n}|_{\text{wall}}=0$). The case B.C 3 has all sidewalls as perfect insulators barring the bottom wall, which is a perfect conductor. Lastly, B.C 4 is a case with all walls as perfect electrical insulators.

B.C 1 is the trivial case where the fluid potential is forced to match the wall potential i.e. $\Psi\big|_{\Omega}=0$ [6]. In this particular case, the electrical properties of bottom wall are inconsequential as the currents are purely horizontal and recirculate within the vertical sidewalls. The current no-outflow conditions in B.C 2, B.C 3 and B.C 4 are crucial for determining the induced electric potentials. Further details regarding the solution of this pure boundary value problem are provided in Sec. I of the supplementary material [22]. The final result for each case are provided below.

First, for B.C 2:

$$\Psi(\text{B.C 2})=\sum_{a,b=0}^{\infty}\grave{\mathcal{B}}_{ab}\left(\frac{1-(-1)^n}{2\,\text{sech}\,[\grave{\xi}_y y]}+\frac{1+(-1)^n}{2\,\text{csch}\,[\grave{\xi}_y y]}\right)\sin\left[\frac{a\pi}{L_x}\left(x+\frac{L_x}{2}\right)\right]\cos\left[\frac{b\pi}{h}(z+h)\right]. \tag{11}$$

Here, only the insulating pair of sidewalls normal to the y-axis, the bottom wall and free surface contribute to the electric potential. The contribution from the non-shallow depth h adds an additional sum to the solution that is absent in the shallow limit [14]. The hyperbolic functions alternate based on the parity of the wave mode n. The sine and cosine terms are attributed to the conducting pair of sidewalls and the insulating boundaries at the bottom and the free surface, respectively. The coefficient $\grave{\mathcal{B}}_{ab}$ is non-zero only for the the condition $a=m$ and the potential $\Psi(\text{B.C 2})=0$ when $m=0$. Therefore, Eq. (11) simplifies to

$$\Psi(\text{B.C 2})=\sum_{b=0}^{\infty}\grave{\mathcal{B}}_{mb}\left(\frac{1-(-1)^n}{2\,\text{sech}\,[\grave{\xi}_y y]}+\frac{1+(-1)^n}{2\,\text{csch}\,[\grave{\xi}_y y]}\right)\sin\left[\frac{m\pi}{L_x}\left(x+\frac{L_x}{2}\right)\right]\cos\left[\frac{b\pi}{h}(z+h)\right], \tag{12}$$

where,

$$\grave{\xi}_y=\pi\sqrt{\frac{m^2}{L_x^2}+\frac{b^2}{h^2}},\quad \grave{\mathcal{B}}_{mb}=\frac{-2A_{mn}\,\omega B_z m\pi(-1)^{n+b}}{\epsilon_b\,\grave{\xi}_y L_x\left(\dfrac{1-(-1)^n}{2\,\text{csch}\,[0.5\grave{\xi}_y L_y]}+\dfrac{1+(-1)^n}{2\,\text{sech}\,[0.5\grave{\xi}_y L_y]}\right)}\frac{h}{(k^2h^2+b^2\pi^2)},\quad \epsilon_r=\begin{cases}2, & r=0\\ 1, & r\neq 0\end{cases}. \tag{13}$$

Second, for B.C 3:

$$\begin{aligned}\Psi(\text{B.C 3})=\sin\left[(z+h)p_h\right]&\left(\frac{\tilde{\mathcal{A}}_0\big((-1)^m-(-1)^n\big)\,\sinh\,[p_h x]}{2p_h\cosh\,[0.5p_hL_x]}-\frac{\tilde{\mathcal{A}}_0\big((-1)^m+(-1)^n\big)\,\cosh\,[p_h x]}{2p_h\sinh\,[0.5p_hL_x]}\right.\\&\left.+\frac{\tilde{\mathcal{B}}_0\big((-1)^n-(-1)^m\big)\,\sinh\,[p_h y]}{2p_h\cosh\,[0.5p_hL_y]}-\frac{\tilde{\mathcal{B}}_0\big((-1)^n+(-1)^m\big)\,\cosh\,[p_h y]}{2p_h\sinh\,[0.5p_hL_y]}\right)\\&+\sum_{\substack{a,b\geq 0\\(a,b)\neq(0,0)\\a\neq n}}^{\infty}\tilde{\mathcal{A}}_{ab}\left(\frac{1-(-1)^m}{2\,\text{sech}\,[\tilde{\xi}_x x]}+\frac{1+(-1)^m}{2\,\text{csch}\,[\tilde{\xi}_x x]}\right)\cos\left[\frac{a\pi}{L_y}\left(y+\frac{L_y}{2}\right)\right]\sin\Big[(2b+1)p_h\big(z+h\big)\Big]\\&+\sum_{\substack{a,b\geq 0\\(a,b)\neq(0,0)\\a\neq m}}^{\infty}\tilde{\mathcal{B}}_{ab}\left(\frac{1-(-1)^n}{2\,\text{sech}\,[\tilde{\xi}_y y]}+\frac{1+(-1)^n}{2\,\text{csch}\,[\tilde{\xi}_y y]}\right)\cos\left[\frac{a\pi}{L_x}\left(x+\frac{L_x}{2}\right)\right]\sin\Big[(2b+1)p_h\big(z+h\big)\Big],\end{aligned} \tag{14}$$

with coefficients,

$$p_h = \frac{\pi}{2h}, \quad \tilde{\xi}_x = \pi\sqrt{\frac{a^2}{L_y^2} + \frac{(1+2b)^2}{4h^2}}, \quad \tilde{\xi}_y = \pi\sqrt{\frac{a^2}{L_x^2} + \frac{(1+2b)^2}{4h^2}}, \quad \epsilon_r = \begin{cases} 2, & r = 0 \\ 1, & r \neq 0 \end{cases}, \tag{15}$$

$$\tilde{\mathcal{A}}_0 = \frac{-4A_{mn}\omega B_z\left((-1)^{m+1} + (-1)^{m+n}\right)}{kL_y}\frac{(2kh + \pi\operatorname{csch}[kh])}{4k^2h^2 + \pi^2}, \tag{16}$$

$$\tilde{\mathcal{B}}_0 = \frac{4A_{mn}\omega B_z\left((-1)^{n+1} + (-1)^{m+n}\right)}{kL_x}\frac{(2kh + \pi\operatorname{csch}[kh])}{4k^2h^2 + \pi^2}, \tag{17}$$

$$\tilde{\mathcal{A}}_{ab} = \frac{8A_{mn}\,\omega B_z n^2\left((-1)^{m+1} + (-1)^{a+m+n}\right)}{\epsilon_a\left(a^2 - n^2\right)\left(\dfrac{1-(-1)^m}{2\operatorname{csch}\left[0.5\tilde{\xi}_x L_x\right]} + \dfrac{1+(-1)^m}{2\operatorname{sech}\left[0.5\tilde{\xi}_x L_x\right]}\right)\tilde{\xi}_x L_y k}\frac{\left(2kh(-1)^b + (1+2b)\pi\operatorname{csch}[kh]\right)}{\left(4k^2h^2 + (1+2b)^2\pi^2\right)}, \tag{18}$$

$$\tilde{\mathcal{B}}_{ab} = \frac{-8A_{mn}\,\omega B_z m^2\left((-1)^{n+1} + (-1)^{a+m+n}\right)}{\epsilon_a\left(a^2 - m^2\right)\left(\dfrac{1-(-1)^n}{2\operatorname{csch}\left[0.5\tilde{\xi}_y L_y\right]} + \dfrac{1+(-1)^n}{2\operatorname{sech}\left[0.5\tilde{\xi}_y L_y\right]}\right)\tilde{\xi}_y L_x k}\frac{\left(2kh(-1)^b + (1+2b)\pi\operatorname{csch}[kh]\right)}{\left(4k^2h^2 + (1+2b)^2\pi^2\right)}. \tag{19}$$

In Eq. (14), the terms containing the coefficients $\tilde{\mathcal{A}}_{ab}$ and $\tilde{\mathcal{B}}_{ab}$ correspond to the insulating boundary conditions of the sidewalls perpendicular to the x- and y-axis accordingly. The constant terms $\tilde{\mathcal{A}}_0$ and $\tilde{\mathcal{B}}_0$ are necessary to satisfy the boundary conditions when $a, b = 0$; a condition where $\tilde{\xi}_x = \tilde{\xi}_y = p_h$ and needs to be separated from the double sums to avoid indeterminate representation of $\tilde{\mathcal{A}}_{ab}$ and $\tilde{\mathcal{B}}_{ab}$.

Lastly, for B.C 4:

$$\begin{aligned}\Psi(\text{B.C } 4) = {}& \ddot{\mathcal{A}}_0 x - \frac{\ddot{\mathcal{A}}_0\left((-1)^n + (-1)^{m+n+1}\right)}{2L_x}\left(x - \frac{L_x}{2}\right)^2 + \ddot{\mathcal{B}}_0 y - \frac{\ddot{\mathcal{B}}_0\left((-1)^m + (-1)^{m+n+1}\right)}{2L_y}\left(y - \frac{L_y}{2}\right)^2 \\ &+ \sum_{\substack{a,b\geq 0 \\ (a,b)\neq(0,0) \\ a\neq n}}^{\infty} \ddot{\mathcal{A}}_{ab}\left(\frac{1-(-1)^m}{2\operatorname{sech}\left[\ddot{\xi}_x x\right]} + \frac{1+(-1)^m}{2\operatorname{csch}\left[\ddot{\xi}_x x\right]}\right)\cos\left[\frac{a\pi}{L_y}\left(y + \frac{L_y}{2}\right)\right]\cos\left[\frac{b\pi}{h}(z+h)\right] \\ &+ \sum_{\substack{a,b\geq 0 \\ (a,b)\neq(0,0) \\ a\neq m}}^{\infty} \ddot{\mathcal{B}}_{ab}\left(\frac{1-(-1)^n}{2\operatorname{sech}\left[\ddot{\xi}_y y\right]} + \frac{1+(-1)^n}{2\operatorname{csch}\left[\ddot{\xi}_y y\right]}\right)\cos\left[\frac{a\pi}{L_x}\left(x + \frac{L_x}{2}\right)\right]\cos\left[\frac{b\pi}{h}(z+h)\right] + C,\end{aligned} \tag{20}$$

with coefficients,

$$\ddot{\xi}_x = \pi\sqrt{\frac{a^2}{L_y^2} + \frac{b^2}{h^2}}, \quad \ddot{\xi}_y = \pi\sqrt{\frac{a^2}{L_x^2} + \frac{b^2}{h^2}}, \quad \epsilon_r = \begin{cases} 2, & r = 0 \\ 1, & r \neq 0 \end{cases}, \tag{21}$$

$$\ddot{\mathcal{A}}_0 = -A_{mn}\frac{\omega B_z\left((-1)^{m+1} + (-1)^{m+n}\right)}{k^2 L_y h}, \qquad \ddot{\mathcal{B}}_0 = A_{mn}\frac{\omega B_z\left((-1)^{n+1} + (-1)^{m+n}\right)}{k^2 L_x h}, \tag{22}$$

$$\ddot{\mathcal{A}}_{ab} = \frac{4A_{mn}\,\omega B_z n^2\left((-1)^{m+b+1} + (-1)^{a+b+m+n}\right)}{\epsilon_a\epsilon_b\left(a^2 - n^2\right)\left(\dfrac{1-(-1)^m}{2\operatorname{csch}\left[0.5\ddot{\xi}_x L_x\right]} + \dfrac{1+(-1)^m}{2\operatorname{sech}\left[0.5\ddot{\xi}_x L_x\right]}\right)\ddot{\xi}_x L_y}\frac{h}{\left(k^2h^2 + b^2\pi^2\right)}, \tag{23}$$

$$\ddot{\mathcal{B}}_{ab} = \frac{-4A_{mn}\,\omega B_z m^2\left((-1)^{n+b+1} + (-1)^{a+b+m+n}\right)}{\epsilon_a\epsilon_b\left(a^2 - m^2\right)\left(\dfrac{1-(-1)^n}{2\operatorname{csch}\left[0.5\ddot{\xi}_y L_y\right]} + \dfrac{1+(-1)^n}{2\operatorname{sech}\left[0.5\ddot{\xi}_y L_y\right]}\right)\ddot{\xi}_y L_x}\frac{h}{\left(k^2h^2 + b^2\pi^2\right)}, \tag{24}$$

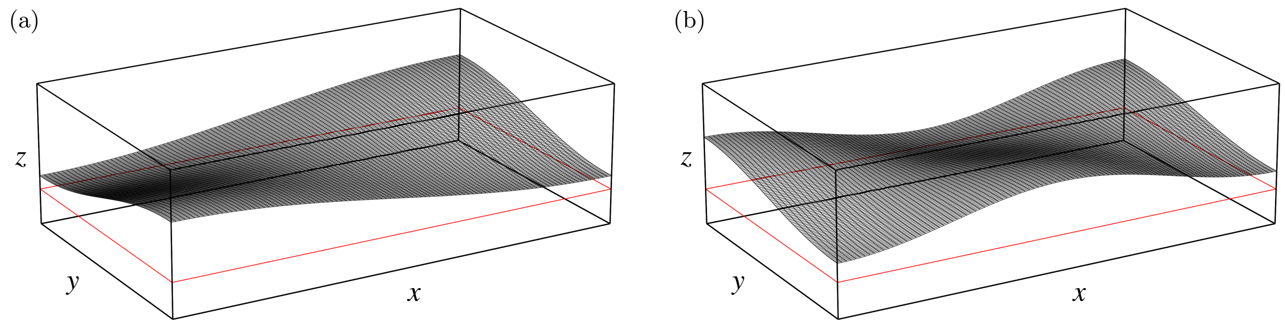


FIG. 2. The free surface wave elevation for a standing wave of mode a) $(1, 1)$ and b) $(2, 1)$ in a rectangular domain. The horizontal cross-section of the volume is highlighted in red, and is the area of interest for the mapping of wave-induced eddy currents.

and gauge $C = 0$, where C is the normalization constraint. The terms containing $\ddot{\mathcal{A}}_0$ and $\ddot{\mathcal{B}}_0$ are to satisfy the boundary condition when $a, b = 0$ in the double sum.

Although Eqs. (12, 14 and 20) represent the induced electric potentials of single-layer conducting fluid systems, they can also be applied to two-layer stratified systems with large conductivity differences (eg., aluminium reduction cells, liquid metal batteries). In these systems, the interface acts like an electrically insulating wall given by the interfacial condition $\partial_z \Psi_i|_{z=0} \approx 0$, where $i = 1, 2$ is the index of the upper and lower layer [19].

C. Eddy current distribution

Since the expanded solutions of the electric potentials are very intricate and the eddy current patterns induced by the wave motions have not yet been analyzed for non-zero wave modes, we want to dive deeper into the underlying physics and examine the solution of the induced current density graphically. For this purpose, we consider a pair of non-zero wave modes $(1, 1)$ and $(2, 1)$ as shown in Fig. 2 and plot the induced current density distribution and streamlines formed at a horizontal plane (outlined in red in Fig. 2) with vertical co-ordinates $z = -h/2$. This is shown in Fig. 3 for all the four boundary conditions. The streamlines have been superimposed on the normalized planar current density distribution using the local maximum. The horizontal induced current densities are given by:

$$\begin{bmatrix} J_x \\ J_y \end{bmatrix} = -\sigma \begin{bmatrix} \Psi_x \\ \Psi_y \end{bmatrix} + \sigma \begin{bmatrix} \phi_y \, B_z \\ -\phi_x \, B_z \end{bmatrix} \tag{25}$$

Figs. 3(a) and 3(b) show the eddy current distribution for waves modes $(1, 1)$ and $(2, 1)$, bounded by perfect conductors as sidewalls (B.C 1) respectively. The solid and dashed lines represent opposite direction of eddy current flow. The pure horizontal currents $\boldsymbol{J} = \sigma(\boldsymbol{u} \times B_z \boldsymbol{e}_z)$, have no boundary corrections and and therefore, recirculate in the wall. When all sidewalls are conducting, the distribution of current density remain the same, regardless of the electrical boundary condition of the bottom wall. In B.C 2, the streamlines close only in the two conducting walls perpendicular to the x-axis, as shown in Figs. 3(c) and 3(d). The eddies closer to the insulating walls tend to close within itself. Figs. 3(e) and 3(f) refer to the eddy current distribution for the B.C 3 boundary configuration and Figs. 3(g) and 3(h) show the same for B.C 4. In both cases, we observe similar eddy current streamlines, but different current density distributions. In B.C 3, the induced currents cannot leave though the sidewalls due to the insulating boundary condition but close through the bottom wall. This can be seen in Figs. 4(a) and 4(b) which shows the current distribution and current streamlines at a vertical cross-section at $y = L_y/2$ for wave modes $(1, 1)$ and $(2, 1)$ respectively. Here, we observe a greater concentration of currents at the bottom wall $z = -h$. Figs. 4(c) and 4(d) show the current distribution at $y = L_y/2$ in the x-z plane for B.C 4 respectively, which is characterized by stronger current densities near the free surface and the redirection of the currents near the insulating bottom wall. It is evident that B.C 3 has higher magnitudes of J_z compared to B.C 4 and, naturally, B.C 2. The difference in current distribution between cases B.C. 3 and B.C. 4, which have the same sidewall boundary conditions except for the bottom wall, is a key result to note when discussing the magnetic damping rate in Sec. III.

By analyzing the solutions in detail, we can deduce an observed rule for the total number of all the eddies that are formed n_{eddy}, as a function of the wave mode indices:

$$n_{\text{eddy}} = (m + 1)\,(n + 1). \tag{26}$$

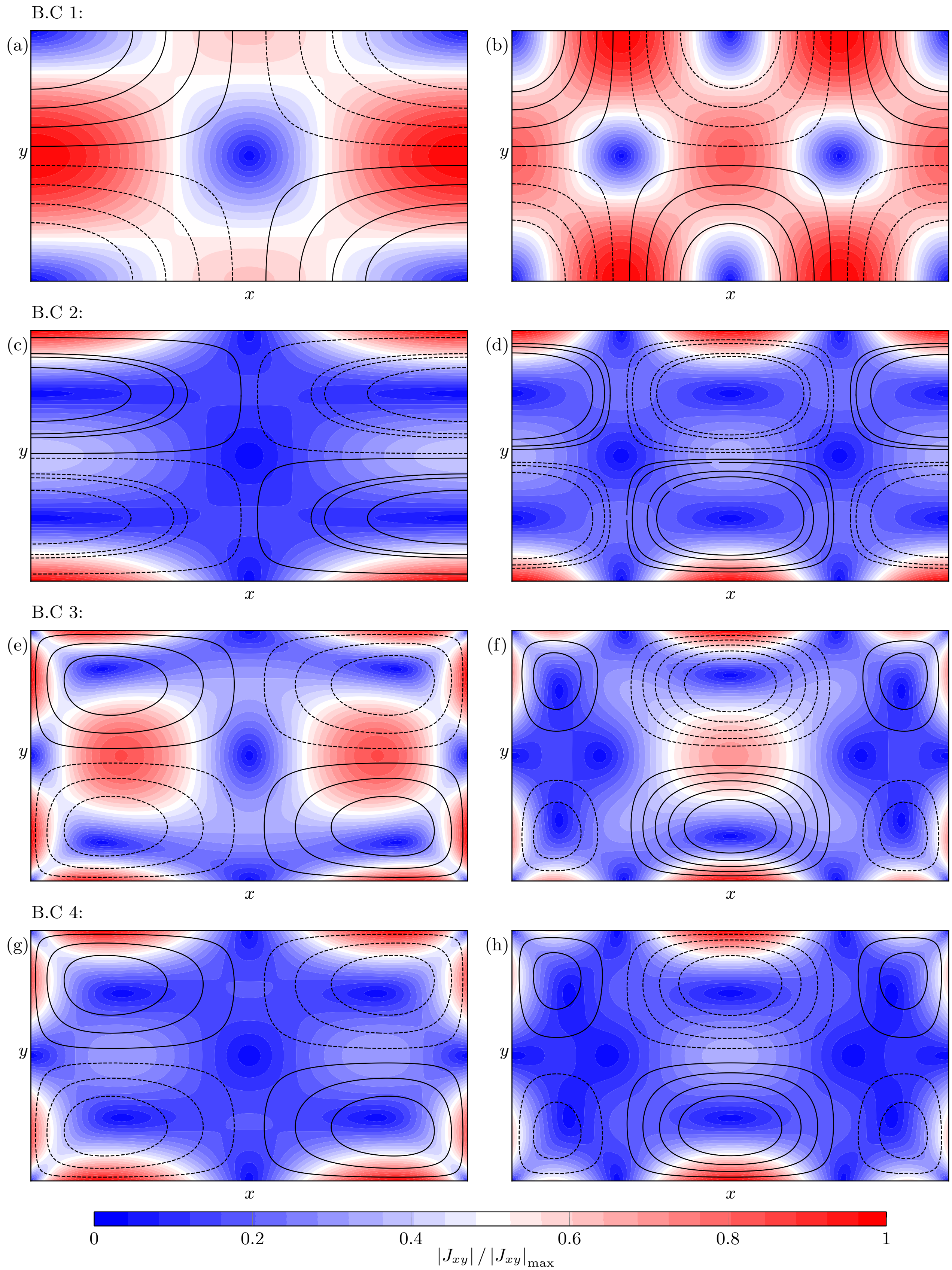


FIG. 3. The induced current streamlines superposed over the normalized current density distribution at a horizontal x-y plane located at $z = -h/2$. Sub-figures a), c), e) and g) pertain to the wave mode $(1,1)$ and b), d), f) and h) pertain to the wave mode $(2,1)$. Solid and dashed lines distinguish opposite current paths.

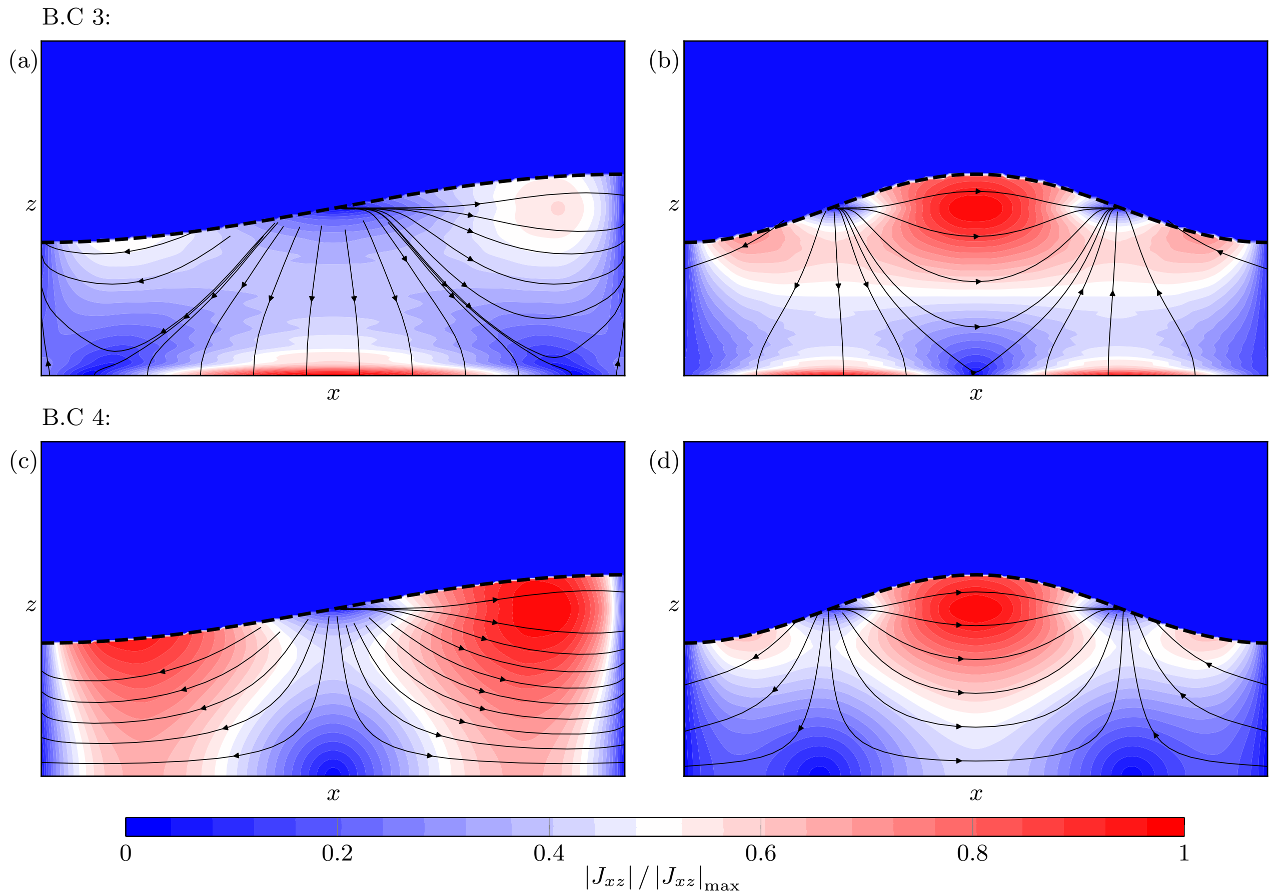


FIG. 4. The normalized induced current density superposed on the streamlines at a cross-section $y = L_y/2$ in the x-z plane. Sub-figures a) and b) show this for wave modes $(1, 1)$ and $(2, 1)$, respectively, for B.C 3; sub-figures c) and d) show the same for B.C 4.

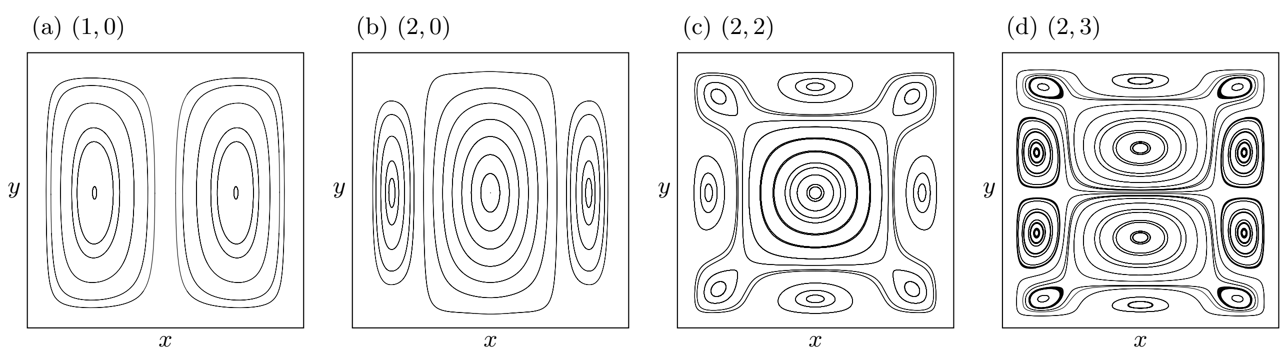


FIG. 5. More examples of induced eddy current streamlines in the horizontal x-y plane for wave modes a) $(1, 0)$, b) $(2, 0)$, c) $(2, 2)$ and d) $(2, 3)$. The total number of induced eddy currents for a given wave mode (m, n) is observed to be equal to $(m + 1)(n + 1)$.

Further examples of eddy current streamlines are given in Fig. 5 for both uni- and bidirectional wave modes pertaining to the B.C 4 case. Wave modes $(1, 0)$, $(2, 0)$, $(2, 2)$ and $(2, 3)$ give rise to 2, 3, 9 and 12 distinct eddies in the horizontal plane respectively. It is evident that the wave modes as well as the boundary conditions clearly have an influence in the complex behavior of the eddy patterns, both in their spatial arrangement and the size distribution.

D. Ohmic dissipation

Having obtained the solutions of Ψ in terms of a series harmonic and hyperbolic functions, we can now solve for $\boldsymbol{J}$ and therefore the damping rate due to Ohmic dissipation, which we denote by λ_{o}. Neglecting the viscosity and surface tension, the mechanical energy balance reads

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{1}{2}\int_V \rho\|\boldsymbol{u}\|^2 \mathrm{d}V\right) + \frac{\mathrm{d}}{\mathrm{d}t}\left(\int_{\mathcal{S}} \frac{1}{2}\rho\, g\eta^2 \,\mathrm{d}\mathcal{S}\right) = -\int_V \frac{1}{\sigma}\boldsymbol{J}^2 \,\mathrm{d}V, \tag{27}$$

which states that the sum of the rate of kinetic energy in the bulk and the potential energy in the free surface is equal to the total energy dissipated. The detailed analytical calculation is provided in Sec. II of the supplementary material [22]. The expression for λ_{o} obtained after accounting for the mean Ohmic loss over a wave cycle, is as follows:

$$\lambda_{\text{o}} = \frac{\mathcal{Q}_{\text{ind}} - \mathcal{Q}_{\text{corr}}}{2K}, \tag{28}$$

where $\mathcal{Q}_{\text{ind}}$ is the dissipation of energy due to induced currents. The expression for $\mathcal{Q}_{\text{ind}}$ for all cases are:

$$\mathcal{Q}_{\text{ind}} = \sigma\int_V \|\boldsymbol{u}\times B_z\boldsymbol{e}_z\|^2 = A_{mn}^2\epsilon_m\epsilon_n\frac{\sigma\omega^2B_z^2(\coth[kh] + kh\,\mathrm{csch}^2\,[kh])}{8k}L_xL_y, \quad \epsilon_r = \begin{cases} 2, & r=0 \\ 1, & r\neq 0 \end{cases}, \tag{29}$$

and $\mathcal{Q}_{\text{corr}}$ is the correction to the dissipation due to insulating/conducting wall boundary conditions, given by the integral

$$\mathcal{Q}_{\text{corr}} = \int_V \boldsymbol{\nabla}\phi\cdot(-\sigma\boldsymbol{\nabla}\Psi\times B_z\boldsymbol{e}_z), \tag{30}$$

which varies depending on the electric potential Ψ. The term K is proportional to the kinetic/potential energy of the standing wave (m,n). It is given by

$$K = 0.25A_{mn}^2\epsilon_m\epsilon_n\rho g L_xL_y. \tag{31}$$

The correction term $\mathcal{Q}_{\text{corr}}$ mainly originates from insulating sidewalls. In a domain with both insulating and conducting sidewalls, the correction terms from each pair of insulating walls are subtracted from the dissipation term if all walls were conducting. In B.C 1, or in a domain with all four sidewalls as perfect conductors, the correction term $\mathcal{Q}_{\text{corr}} = 0$, as the current densities close in the walls (see Fig. 3(a) and 3(b)).

In B.C 2, having one pair of insulating sidewalls results in:

$$\mathcal{Q}_{\text{corr}}(\text{B.C 2}) = A_{mn}\left[\sum_{b=0}^{\infty}\frac{-\grave{\mathcal{B}}_{mb}\,\sigma\omega B_z\,m\pi(-1)^{n+b}}{\left(\dfrac{1-(-1)^n}{2\cosh\left[0.5\grave{\xi}_yL_y\right]} + \dfrac{1+(-1)^n}{2\sinh\left[0.5\grave{\xi}_yL_y\right]}\right)}\frac{h^2}{(k^2h^2+b^2\pi^2)}\right], \tag{32}$$

as the pair of conducting walls parallel to the y-axis do not contribute to the correction term; which is why $\grave{\mathcal{Q}}_{\text{corr}} = 0$ for a unidirectional wave along the y-axis ($m=0$, $y\neq 0$). The terms $\grave{\mathcal{B}}_{mb}$ and $\grave{\xi}_y$ are given in Eq. (13).

The correction term for B.C 3 is

$$\mathcal{Q}_{\rm corr}(\text{B.C }3) = A_{mn}\Bigg[\tilde{\mathcal{A}}_0\sigma\omega B_z\left(\frac{\tilde{\Delta}_{m,n}\tanh\left[0.5p_hL_x\right]+\tilde{\tilde{\Delta}}_{m,n}\coth\left[0.5p_hL_x\right]}{2\pi(m^2h^2+0.25L_x^2)}\right)\frac{h^2L_x^2(\pi\operatorname{csch}[kh]+2kh)}{k(4k^2h^2+\pi^2)}$$
$$-\tilde{\mathcal{B}}_0\sigma\omega B_z\left(\frac{\tilde{\Delta}_{n,m}\tanh\left[0.5p_hL_y\right]+\tilde{\tilde{\Delta}}_{n,m}\coth\left[0.5p_hL_y\right]}{2\pi(n^2h^2+0.25L_y^2)}\right)\frac{h^2L_y^2(\pi\operatorname{csch}[kh]+2kh)}{k(4k^2h^2+\pi^2)}$$
$$+\sum_{\substack{a,b\geq 0\\(a,b)\neq(0,0)\\a\neq n}}^{\infty}\frac{4\tilde{\mathcal{A}}_{ab}\,\omega\sigma B_z\,(a^2m^2\pi^2+\tilde{\xi}_x^2n^2L_x^2)\Big((-1)^{m+1}+(-1)^{a+m+n}\Big)}{(a^2-n^2)(m^2\pi^2+\tilde{\xi}_x^2L_x^2)\left(\dfrac{1-(-1)^m}{2\cosh\left[0.5\tilde{\xi}_xL_x\right]}+\dfrac{1+(-1)^m}{2\sinh\left[0.5\tilde{\xi}_xL_x\right]}\right)}\frac{h\Big((1+2b)\pi\operatorname{csch}[kh]+2kh(-1)^b\Big)}{k\Big(4k^2h^2+(1+2b)^2\pi^2\Big)}$$
$$-\sum_{\substack{a,b\geq 0\\(a,b)\neq(0,0)\\a\neq m}}^{\infty}\frac{4\tilde{\mathcal{B}}_{ab}\,\omega\sigma B_z\,(a^2n^2\pi^2+\tilde{\xi}_y^2m^2L_y^2)\Big((-1)^{n+1}+(-1)^{a+m+n}\Big)}{(a^2-m^2)(n^2\pi^2+\tilde{\xi}_y^2L_y^2)\left(\dfrac{1-(-1)^n}{2\cosh\left[0.5\tilde{\xi}_yL_y\right]}+\dfrac{1+(-1)^n}{2\sinh\left[0.5\tilde{\xi}_yL_y\right]}\right)}\frac{h\Big((1+2b)\pi\operatorname{csch}[kh]+2kh(-1)^b\Big)}{k\Big(4k^2h^2+(1+2b)^2\pi^2\Big)}\Bigg], \tag{33}$$

where,

$$\tilde{\Delta}_{q,r}=(1+(-1)^q)(1-(-1)^r)^2,\qquad \tilde{\tilde{\Delta}}_{q,r}=(-1+(-1)^q)(1-(-1)^r)^2. \tag{34}$$

The constants $\tilde{\xi}_x$ and $\tilde{\xi}_y$ are given in Eq. (15); and the coefficients $\tilde{\mathcal{A}}_0$, $\tilde{\mathcal{B}}_0$, $\tilde{\mathcal{A}}_{ab}$ and $\tilde{\mathcal{B}}_{ab}$ are given in Eqs. (16, 17, 18 and 19) respectively.

Similarly, in B.C 4, the correction term takes the following form:

$$\mathcal{Q}_{\rm corr}(\text{B.C }4) = A_{mn}\Bigg[-\ddot{\mathcal{A}}_0\sigma\omega B_zL_x\frac{(-1+(-1)^n)}{k^2}\,\ddot{\Delta}_m+\ddot{\mathcal{B}}_0\sigma\omega B_zL_y\frac{(-1+(-1)^m)}{k^2}\,\ddot{\Delta}_n$$
$$+\sum_{\substack{a,b\geq 0\\(a,b)\neq(0,0)\\a\neq n}}^{\infty}\frac{2\ddot{\mathcal{A}}_{ab}\sigma\omega B_z\,(a^2m^2\pi^2+\ddot{\xi}_x^2n^2L_x^2)\Big((-1)^{b+m+1}+(-1)^{a+b+m+n}\Big)}{(a^2-n^2)(m^2\pi^2+\ddot{\xi}_x^2L_x^2)\left(\dfrac{1-(-1)^m}{2\cosh\left[0.5\ddot{\xi}_xL_x\right]}+\dfrac{1+(-1)^m}{2\sinh\left[0.5\ddot{\xi}_xL_x\right]}\right)}\frac{h^2}{(k^2h^2+b^2\pi^2)}$$
$$-\sum_{\substack{a,b\geq 0\\(a,b)\neq(0,0)\\a\neq m}}^{\infty}\frac{2\ddot{\mathcal{B}}_{ab}\sigma\omega B_z\,(a^2n^2\pi^2+\ddot{\xi}_y^2m^2L_y^2)\Big((-1)^{b+n+1}+(-1)^{a+b+m+n}\Big)}{(a^2-m^2)(n^2\pi^2+\ddot{\xi}_y^2L_y^2)\left(\dfrac{1-(-1)^n}{2\cosh\left[0.5\ddot{\xi}_yL_y\right]}+\dfrac{1+(-1)^n}{2\sinh\left[0.5\ddot{\xi}_yL_y\right]}\right)}\frac{h^2}{(k^2h^2+b^2\pi^2)}\Bigg], \tag{35}$$

where,

$$\ddot{\Delta}_m=\begin{cases}1, & m=0,n\neq 0\\ \dfrac{(-1)^n\big(1-(-1)^m\big)^2}{m^2\pi^2}, & m,n\neq 0\end{cases},\quad \ddot{\Delta}_n=\begin{cases}1, & n=0,m\neq 0\\ \dfrac{(-1)^m\big(1-(-1)^n\big)^2}{n^2\pi^2}, & m,n\neq 0\end{cases}. \tag{36}$$

In Eq. (35), the coefficients $\ddot{\mathcal{A}}_0$, $\ddot{\mathcal{B}}_0$ are given in Eq. (22), and $\ddot{\mathcal{A}}_{ab}$ and $\ddot{\mathcal{B}}_{ab}$ are given in Eqs. (23 and 24) respectively.

E. Hartmann and Shercliff boundary layers

In addition to Ohmic dissipation in the bulk, the effect of the magnetic field on the boundary must be taken into consideration. In an electrically conducting fluid system, the presence of a magnetic field alters the standard hydrodynamic boundary layers depending on the field intensity. The velocity profile at the walls have two distinct layers, called the Hartmann and Shercliff layers. Hartmann boundary layers are formed at surfaces perpendicular to the direction of the applied magnetic field, whereas Shercliff layers are formed parallel to them. From literature, we know that the thickness of the Hartmann layers scale inversely with Ha ($\delta_{\mathrm{Ha}} \sim \mathrm{Ha}^{-1}$) [5, 23, 24]. This implies that increasing the magnetic field strength reduces the overall thickness while increasing overall dissipation at the wall. Shercliff layers, unlike Hartmann layers do not directly contribute in braking the fluid velocity but redistribute the induced currents and fluid momentum throughout the system. They are usually thicker than the Hartmann layers as they scale inversely to the square root of Ha ($\delta_{\mathrm{Sh}} \sim \mathrm{Ha}^{-0.5}$) [5].

In the context of our current model, the Hartmann layer forms only at the bottom wall of the domain. The Shercliff layers form at the four sidewalls of the rectangular domain, which are parallel to the applied magnetic field. In principle, these layers are the result of the Stokes boundary layer transitioning under the influence of the magnetic field. They are a combination of the conventional Hartmann or Shercliff layers with the respective oscillating Stokes boundary layer [14]. Therefore we can follow the methodology of Keulegan [25] in resolving the Stokes boundary layers for free surface waves in a rectangular domain. We can then incorporate the effects of Lorentz forces due to induced currents [6]. The inviscid formulation of the velocity $\boldsymbol{u} = \boldsymbol{\nabla}\phi$ already satisfies the condition of impermeability at the boundary. However, in order to model the tangential velocity and stress at the boundary, we need to enforce the following corrections to the potential wave solutions of velocity and pressure:

$$\boldsymbol{u} = \left[\boldsymbol{\nabla}\phi + \bar{\boldsymbol{u}}\,\mathrm{e}^{\mathrm{i}\omega t}\right]\mathrm{e}^{-\lambda_{\mathrm{b}} t}, \tag{37a}$$

$$p = \left[-\rho(\mathrm{i}\omega)\phi + \bar{p}\,\mathrm{e}^{\mathrm{i}\omega t}\right]\mathrm{e}^{-\lambda_{\mathrm{b}} t}, \tag{37b}$$

with $\bar{\boldsymbol{u}}$ and $\bar{p}$ as the corrections to the potential flow velocity and pressure respectively. The term λ_{b} in the exponent is the damping rate of the standing wave solely due to boundary layer effects. Considering only the tangential velocities $\bar{\boldsymbol{u}}_{\perp}$ at the respective boundaries and the damping force by virtue of induction, the Navier-Stokes momentum equation [Eq. (3a)] reduces to

$$\mathrm{i}\omega\bar{\boldsymbol{u}}_{\perp} = -\frac{1}{\rho}\boldsymbol{\nabla}_{\perp}\bar{p} + \nu\frac{\partial^2\bar{\boldsymbol{u}}_{\perp}}{\partial\zeta^2} - \frac{\sigma B_z^2}{\rho}\boldsymbol{u}_b, \tag{38}$$

when considering only the leading order terms. In Eq. (38), ζ is the spatial coordinate normal to the wall that reduces the boundary layer corrections to zero away from the wall:

$$\zeta = \begin{cases} z+h, & \text{near } z=-h, \\ \dfrac{L_x}{2}-x, & \text{near } x = L_x/2, \qquad x+\dfrac{L_x}{2}, \quad \text{near } x=-L_x/2 \\ \dfrac{L_y}{2}-y, & \text{near } y=L_y/2, \qquad y+\dfrac{L_y}{2}, \quad \text{near } y=-L_y/2, \end{cases}, \tag{39}$$

and $\boldsymbol{u}_b$ is the velocity in the boundary layer. In the leading order, the normal part of the mass and momentum balance is zero i.e. $\partial_\zeta\bar{\boldsymbol{u}}\cdot\boldsymbol{n} = 0$ and $\bar{p} = 0$ respectively. The tangential velocities decay exponentially away from the boundary, $\bar{\boldsymbol{u}}_{\perp} \sim \mathrm{e}^{-\zeta/\delta_{\mathrm{tot}}}$, where δ_{tot} is the total thickness of the boundary layer. The inverse of δ_{tot} signifies the sharpness of the velocity gradient near the boundary. In context to the Hartmann layer, the velocity at the boundary scales as $\boldsymbol{u}_b \sim \bar{\boldsymbol{u}}_{\perp}$. We can therefore solve for δ_{tot} in Eq. (38) to obtain

$$\delta_{\mathrm{tot}}(\mathrm{Ha}) = \left[\mathrm{Re}\sqrt{\frac{\mathrm{i}\omega}{\nu}+\frac{1}{\tau\nu}}\right]^{-1} = \frac{\sqrt{2\tau\nu}}{\sqrt{1+\sqrt{1+\tau^2\omega^2}}} = \left(\frac{1}{2}\left(\delta_{\mathrm{Ha}}^{-2} + \left(\delta_{\mathrm{Ha}}^{-4} + 4\delta_{\mathrm{v}}^{-4}\right)^{0.5}\right)\right)^{-0.5}, \tag{40}$$

which is the total boundary layer thickness at the wall normal to the magnetic field. Here τ is denoted as the characteristic damping time given by

$$\tau = \frac{\rho}{\sigma B_z^2}. \tag{41}$$

Eq. (40) is a combination of both the viscous ($\delta_{\mathrm{v}} = \sqrt{2\nu/\omega}$) and the Hartmann boundary ($\delta_{\mathrm{Ha}} = \sqrt{\tau\nu}$) layers. However, there is no straight forward equation that can be used to calculate the expression for the Shercliff boundary layer. But we can approximate the total thickness using the conservation of currents. In a given fluid layer, the currents close as they pass through the horizontal Hartmann layers of thickness δ_{Ha} and the vertical Shercliff layers over the entire fluid depth h. Therefore, we write the following relation:

$$\sigma\bar{u}_{\perp}B_z\delta_{\mathrm{Ha}} \sim \sigma u_b B_z h \Longrightarrow u_b \sim \frac{\delta_{\mathrm{Ha}}}{h}\,\bar{u}_{\perp}. \tag{42}$$

TABLE II. Cell parameters and material properties from Bojarevičs *et al.* [12].

Length of cell (L_x):	0.154 m		
Breadth of cell (L_y):	0.094 m		
Properties of GaInSn (Ω)			
Density (ρ):	$6363\,\mathrm{kg\,m^{-3}}$	Electrical conductivity (σ):	$3.31\times10^{6}\,\mathrm{S\,m^{-1}}$
Kinematic viscosity (ν):	$3.48\times10^{-7}\,\mathrm{m^2\,s^{-1}}$	Layer depth (h):	0.049 m

In Eq. (42), the current density in the Hartmann layer scales with the current density in the Shercliff layer over the fluid height. Substituting Eq. (42) in (38), we get the total boundary layer thickness at the sidewalls parallel to the magnetic field:

$$\delta_{\mathrm{tot}}(\mathrm{Sh}) \approx \left[\mathrm{Re}\sqrt{\frac{\mathrm{i}\omega}{\nu}+\frac{1}{h\sqrt{\tau\nu}}}\right]^{-1} = \frac{\sqrt{2h\sqrt{\tau\nu}}}{\sqrt{1+\sqrt{1+\dfrac{\tau h^2\omega^2}{\nu}}}} = \left(\frac{1}{2}\left(\delta_{\mathrm{Sh}}^{-2}+\left(\delta_{\mathrm{Sh}}^{-4}+4\delta_{\mathrm{v}}^{-4}\right)^{0.5}\right)\right)^{-0.5}, \tag{43}$$

where the conventional Shercliff layer scales with the square root of the Hartmann layer thickness, $\delta_{\mathrm{Sh}} \sim \sqrt{h\,\delta_{\mathrm{Ha}}}$. Therefore, the tangential velocity corrections in the Hartmann and Shercliff layers are

$$\bar{\boldsymbol{u}}_{\perp}(\mathrm{Ha}) = -\boldsymbol{\nabla}\phi \mathrm{e}^{-\sqrt{\frac{\mathrm{i}\omega}{\nu}+\frac{1}{\tau\nu}}\zeta}, \qquad \bar{\boldsymbol{u}}_{\perp}(\mathrm{Sh}) = -\boldsymbol{\nabla}\phi \mathrm{e}^{-\sqrt{\frac{\mathrm{i}\omega}{\nu}+\frac{1}{h\sqrt{\tau\nu}}}\zeta}, \tag{44}$$

which contribute to the damping rates of a standing wave solely due to the dissipation of energy in the Hartmann-viscous layer and the Shercliff-viscous layer respectively. We denote the damping rates as λ_{Ha} and the λ_{Sh} respectively.

To illustrate more clearly the effect of the Hartmann and Shercliff layers on the Stokes velocity profile, we calculate the boundary layer velocities in a single-layer liquid metal system from Bojarevičs *et al.* [12]. We consider a free surface wave motion of mode $(1,0)$. The system is threaded by a constant vertical DC magnetic field of magnitude B_z. The properties of the materials are provided in Table II. The velocity profiles from the bottom boundary to the free surface at different Hartmann numbers are presented in Fig. 6(a). The velocities are plotted along the line $x = L_x/4$ in the x-z plane, with $\mathrm{Ha} = B_z L_x\sqrt{\sigma/(\rho\nu)}$. Here, the term $\boldsymbol{u}_T^* = (\boldsymbol{u}_x + \bar{\boldsymbol{u}}_x)/\boldsymbol{u}_x$ is the non-dimensionalized total to bulk velocity ratio. In the absence of the magnetic field ($\mathrm{Ha}=0$), the boundary reverts to the classical Stokes boundary layer with thickness $\delta_{\mathrm{tot}} = \sqrt{2\nu/\omega}$, which is highlighted in blue. The boundary layer thickness at $\mathrm{Ha}=3000$ (highlighted in red) is minuscule in comparison to the case when $\mathrm{Ha}=0$, as δ_{tot} varies inversely with Ha. At $z=-h$, the velocity ratio follows the Stokes profile where the correction cancels out the free stream velocity at the boundary. This is followed by the characteristic 'velocity overshoot', where the local velocity exceeds the bulk velocity momentarily and then converges to the bulk velocity after. This is due to the no-slip constraint enforced at the wall, which does not cause the corrections to fade monotonically but act like a damped oscillator (emphasized by the presence of a cosine term in the velocity) that approaches equilibrium [26]. As the Hartmann number increases, the overshoots die down and are replaced by sharper velocity gradients and the boundary layer shrinks to Eq. (40).

Fig. 6(b) shows the magnified view of velocity profile along the line $z=-h/2$ in the x-z plane, near the sidewall at $x=-L_x/2$. When $\mathrm{Ha}=3000$, we observe that the Shercliff layer is identical to the conventional viscous sub layer ($\mathrm{Ha}=0$), and is significantly thicker than the Hartmann layer at the bottom. The velocity ratio $\boldsymbol{u}_T^* = (\boldsymbol{u}_z + \bar{\boldsymbol{u}}_z)/\boldsymbol{u}_z$ in this case. The Stokes 'overshoot' is noticeable for all cases of Ha considered; however its impact on the sidewall to sidewall velocity profile is negligible in the given range. A significant change in the curve would only occur at extremely high and unrealistic values of Ha.

From the tangential velocities given in Eq. (44), we can now determine the total dissipation in the respective boundary layers. To do so, we use the following mechanical energy balance:

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{1}{2}\int_V \rho\|\boldsymbol{u}\|^2\mathrm{d}V\right) + \frac{\mathrm{d}}{\mathrm{d}t}\left(\int_{\mathcal{S}}\frac{1}{2}\rho\, g\eta^2\,\mathrm{d}\mathcal{S}\right) = -2\rho\nu\int_V \boldsymbol{\varepsilon} : \boldsymbol{\varepsilon}\,\mathrm{d}V, \tag{45}$$

which, like Eq. (27) states that the sum of rate of change of kinetic and potential energy of the wave is equal to the total dissipation at the boundary expressed in terms of the strain rate tensor $\boldsymbol{\varepsilon} = 0.5(\boldsymbol{\nabla}\bar{\boldsymbol{u}}_{\perp} + \boldsymbol{\nabla}\bar{\boldsymbol{u}}_{\perp}^{\mathrm{T}})$. Expanding the right hand side of Eq. (45) as per Lamb [27], we substitute the tangential velocities given in Eq. (44) to determine the damping rate due to the respective Hartmann and Shercliff layers. The calculations pertaining to this are provided

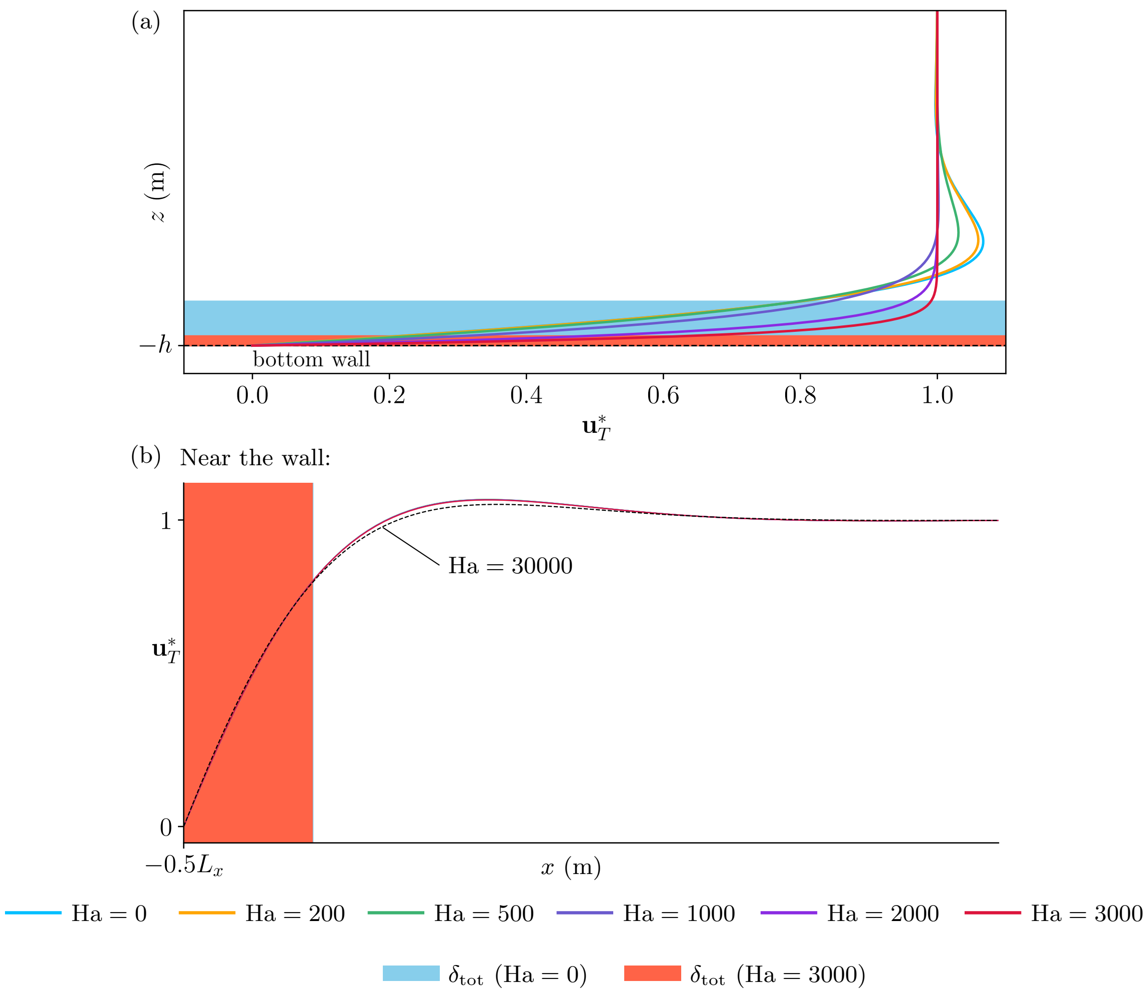


FIG. 6. The Stokes velocity profile in the x-z plane for different Ha near the a) bottom wall along $x = L_x/4$, and b) near the sidewall ($-0.5L_x$) along $z = -h/2$. A quasi-2D unidirectional wave mode $(1, 0)$ is considered. The blue shaded region quantifies the viscous layer thickness whereas the red region shows the total boundary thickness including Hartmann and Shercliff layers in a) and b) respectively.

in Sec. III of the supplementary material [22]. We find the damping contribution due to the Hartmann layer at the bottom to be

$$\lambda_{\text{Ha}} = \frac{\nu k}{2\sqrt{2}} \frac{1}{\sqrt{\tau\nu}} \left(\frac{\sqrt{1+\sqrt{1+\tau^2\omega^2}}}{\sinh{[kh]}\cosh{[kh]}} \right), \tag{46}$$

and

$$\lambda_{\text{Sh}} = \frac{\nu}{\sqrt{2}\,k} \frac{\sqrt{1+\sqrt{1+\dfrac{\omega^2 h^2 \tau}{\nu}}}}{L_x L_y \sqrt{h\sqrt{\tau\nu}}} \left(\left[\frac{h(n^2\pi^2 - k^2 L_y^2)}{\epsilon_m\, L_y \sinh[kh] \cosh{[kh]}} + \frac{(n^2\pi^2 + k^2 L_y^2)}{\epsilon_m L_y k} \right.\right.$$
$$\left.\left. + \frac{h(m^2\pi^2 - k^2 L_x^2)}{\epsilon_n\, L_x \sinh{[kh]}\cosh{[kh]}} + \frac{(m^2\pi^2 + k^2 L_x^2)}{\epsilon_n L_x k} \right] \right), \tag{47}$$

at the sidewalls due to the formation of the Shercliff layer. In Eq. (47), the terms with ϵ_m and ϵ_n correspond to the damping contributions from the sidewalls normal to the x- and y-axis accordingly. The total damping rate due to the boundary layer is given by

$$\lambda_{\text{b}} = \lambda_{\text{Ha}} + \lambda_{\text{Sh}}. \tag{48}$$

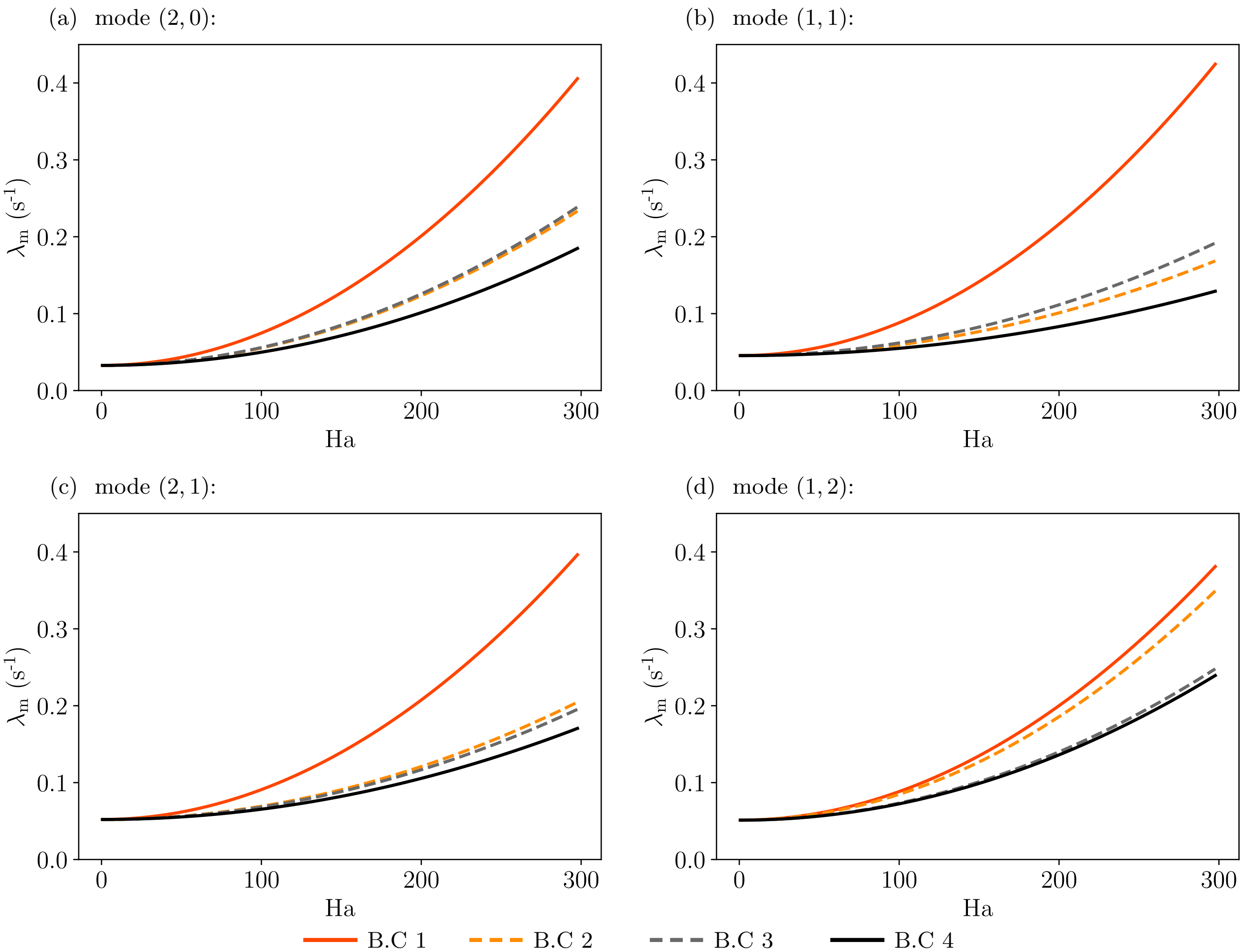


FIG. 7. The analytical magnetic damping rates at different Ha of a single-layer system with different electrical boundary condition mentioned in Table I. The wave modes considered are a) (2, 0), b) (1, 1), c) (2, 1) and d) (1, 2). The cell dimensions and material properties of the fluids are given in Table II.

From Fig. 6, we can infer that, at the current range of Ha, boundary layer dissipation is dominated by the Hartmann layers. The influence of the Shercliff layer in the sidewalls is found to be negligible within the range of Ha considered. The Shercliff boundary layer has nearly indistinguishable influence on the velocity gradients even at extremely high values of $\mathrm{Ha} = 3000$, with only a slight effect seen at $\mathrm{Ha} = 30000$ ($B_z = 5\,\mathrm{T}$). Such high value of Ha is unrealistic for the industrial and experimental models specific to this study. This implies that viscous contributions mainly dictate the damping rate at the sidewalls.

At the shallow depths in the absence of a magnetic field, the bottom and sidewall damping contributions given in Eqs. (46 and 47) reduce to the viscous damping rate formulations presented in the works of Keulegan [25] and further elaborated by Faltinsen and Timokha [28].

III. MAGNETIC DAMPING

The total magnetic damping rate due to both the Ohmic dissipation given in Eq. (28) and the boundary layer given in Eq. (48) is equal to the sum of the damping rates due to ohmic dissipation and boundary effects:

$$\lambda_{\mathrm{m}} = \lambda_{\mathrm{o}} + \lambda_{\mathrm{b}}. \tag{49}$$

To show how the electrical boundary conditions affects magnetic induction, we calculate the magnetic damping rate of a system similar to Bojarevičs *et al.* [12] (refer to Table II for material properties). We calculate the magnetic damping rate at different Ha of a uni- and bidirectional free surface wave in four separate domains with the aforementioned boundary conditions. Figs. 7(a), 7(b), 7(c) and 7(d) represent the damping rates of the wave modes $(2,0)$, $(1,1)$, $(2,1)$ and $(1,2)$ within a range $0 \leq \mathrm{Ha} \leq 300$, accordingly. The initial vertical offset at $B_z = 0\,\mathrm{mT}$ is due to the viscous dissipation at the walls. The damping behavior of B.C 1 and 4 is as expected, with current recirculation in the walls aiding and recirculation within the walls reducing Ohmic dissipation. The conducting sidewalls in B.C 1 provides a low-resistance bypass for the electric currents, whereas in B.C 4, the induced currents are forced to go through the highly resistive shear layer near the walls. This is known to reduce the total current flow in the fluid. The mixed boundary cases of B.C 2 and B.C 3 show contrasting behavior depending on the wave mode. In B.C 2, the configuration of the wave with respect to the orientation of the conducting plates plays a big role in determining the damping rate. A larger mode index in the y-direction ($n > m$) results in increased eddy current flow into the conducting plates oriented normal to the x-axis, thereby increasing the damping rate. Conversely, the damping rate reduces when $m > n$. This can be seen in Figs.7(c) and 7(d) in relation to B.C 2. In B.C 3, the presence of a conducting bottom wall increases the overall Ohmic and boundary dissipation compared to an insulating bottom wall. This is in conjunction with the experimental observations of Domínguez-Lozoya *et al.* [29], who had a similar boundary configuration for swirling flows. Moreover, we also note that the damping curves of B.C 2, B.C 3 and B.C 4 converge to that of B.C 1 for very large wave modes, for all B_z values.

From the results presented above, it is evident that the nature of standing waves, uni- or bidirectional, the electrical boundary condition and its orientation with respect to the wave in some cases, along with the dimensions of the domain, affects the way in which the electric currents complete their paths within the medium and thus, determines the extent of magnetic damping.

A. Verification with Sreenivasan *et al.* [14]

We now independently validate our formula for magnetic dissipation with the experimental results and analytical model presented by Sreenivasan *et al.* [14], which pertains to the decay of free surface waves in shallow rectangular geometries. Table III provides details on the physical parameters and material properties used for the study. Sreenivasan *et al.* [14] have dedicated an entire study to the magnetic attenuation of standing gravity waves in rectangular containers and measured the decay rates of surface waves in liquid mercury that were exposed to vertical magnetic fields of up to 1300 mT. Moreover, the authors provide a theoretical solution for magnetic damping times that takes into account bulk Ohmic dissipation, viscous side wall dissipation and Hartmann layer dissipation (at the bottom). The damping times are given by $\lambda_\mathrm{m}^{-1} = (\lambda_\mathrm{o} + \lambda_\mathrm{b})^{-1}$, also expressed as the inverse of the magnetic damping rate. Their model was the first to properly address the delicate problem of electrically insulating walls (B.C 4), but was only formulated for the first unidirectional wave mode ($m = 1, n = 0$) using the shallow-water approximation. Fig. 8 shows the total damping time as a function of the Hartmann number, which in this context is given by $\mathrm{Ha} = B_z L_y \sqrt{\sigma/(\rho\nu)}$. At the shallow water limit ($kh \ll 1$), the damping rate due to the viscous-Hartmann layer given in Eq. (46) transforms to

$$\lambda_\mathrm{Ha} = \frac{\nu}{2\sqrt{2}\,h}\frac{1}{\sqrt{\tau\nu}}\sqrt{1+\sqrt{1+\tau^2\omega^2}}. \tag{50}$$

Even though Sreenivasan *et al.* [14] does not consider dissipation at the Shercliff layers, we take this opportunity to formulate the damping rate

$$\lambda_\mathrm{Sh} = \frac{\nu}{\sqrt{2}\,L_y}\frac{1}{\sqrt{h\sqrt{\tau\nu}}}\sqrt{1+\sqrt{1+\frac{\tau}{\nu}h^2\omega^2}} \approx \frac{\sqrt{\omega\nu}}{\sqrt{2}\,L_y}, \tag{51}$$

which is approximately equal to the hydrodynamic viscous damping rate at the sidewalls. Eqs. (29, 35, 50 and 51) together coincide with the Ohmic dissipation, Hartmann and viscous layer contributions from Eq. 37 given in [14]. This is shown in Figs. 8(a) and 8(b), where we find good agreement between both theories and experiments at two different layer depths: 22 and 35 mm respectively.

IV. THE MAGNETIC DAMPING EXPERIMENT

We now compare our analytical magnetic damping model from Sec II D using a similar experimental setup from Hegde *et al.* [18]. As in [14], we excited the free surface of a single-layer system and then proceeded to measure the rate of decay of the amplitude. We operated at a significantly lower range of $100 < \mathrm{Ha} < 500$, in contrast to Sreenivasan

TABLE III. Cell parameters and material properties from Sreenivasan *et al.* [14].

Length of cell (L_x):	0.15 m	Wave mode (m, n):	$(1, 0)$
Breadth of cell (L_y):	0.04 m	Boundary condition:	Perfectly insulating walls. (similar to B.C 4)
Properties of Mercury (Ω)			
Density (ρ):	$13\,546\,\mathrm{kg\,m^{-3}}$	Electrical conductivity (σ):	$1 \times 10^{6}\,\mathrm{S\,m^{-1}}$
Kinematic viscosity (ν):	$1.15 \times 10^{-7}\,\mathrm{m^2\,s^{-1}}$	Layer depths (h):	0.022 m, 0.035 m

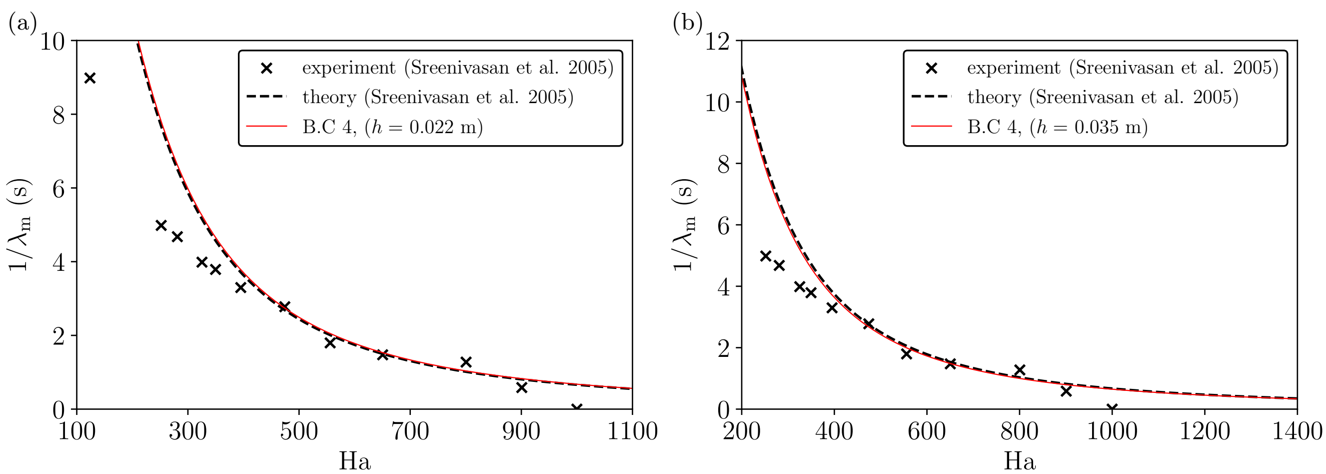


FIG. 8. Comparison of the magnetic damping time which is the inverse of the magnetic damping rate $\lambda_{\mathrm{m}} = \lambda_{\mathrm{o}^*} + \lambda_{\mathrm{v}^*}$, with experimental measurements and theoretical predictions of [14] as a function of the Hartmann number Ha at two different layer depths: a) $h = 0.022\,\mathrm{m}$ and b) $h = 0.035\,\mathrm{m}$ of the liquid metal. The wave mode of the free surface wave was $(1, 0)$.

et al. [14]. A standing wave of mode $(1, 0)$ was magnetohydrodynamically excited using horizontal, externally fed currents. Exciting higher wave modes posed a challenge with the current setup, as these modes rely on the formation of distinct excess current loops within the liquid metal. This can only be achieved by vertically injecting the input current, which is not possible with the current configuration. Of the four different cases covered in the study, only B.C 1 and B.C 4 were realised in the experiment. The experimental setup and results are given in Sec. IV A and IV B respectively.

A. The setup

The specifications of the experimental setup are provided in Table IV. The wave tank was made out of polymethyl methacrylate (PMMA) with inner dimensions of $0.255 \times 0.255 \times 0.1\,\mathrm{m}^3$. The walls were 0.008 m thick. Each of the

TABLE IV. Material properties of the working fluid and system parameters used in the magnetic damping experiment.

Length of tank (L_x):	0.255 m	Wave mode (m, n):	$(1, 0)$
Breadth of tank (L_y):	0.255 m	Boundary condition:	i) Copper plates on sidewalls (B.C 1).
Height of tank (H):	0.1 m		ii) Stainless steel plates on sidewalls (B.C 4)
Properties of GaInSn (Ω)			
Density (ρ):	$6345\,\mathrm{kg\,m^{-3}}$	Electrical conductivity (σ):	$3.2 \times 10^{6}\,\mathrm{S\,m^{-1}}$
Kinematic viscosity (ν):	$3.4 \times 10^{-7}\,\mathrm{m^2\,s^{-1}}$	Layer depth (h):	0.03 m
Properties of copper			
Size:	$0.23 \times 0.1 \times 0.002\,\mathrm{m^3}$	Electrical conductivity:	$58 \times 10^{6}\,\mathrm{S\,m^{-1}}$
Properties of stainless steel			
Size:	$0.23 \times 0.1 \times 0.002\,\mathrm{m^3}$	Electrical conductivity:	$1.1 \times 10^{6}\,\mathrm{S\,m^{-1}}$

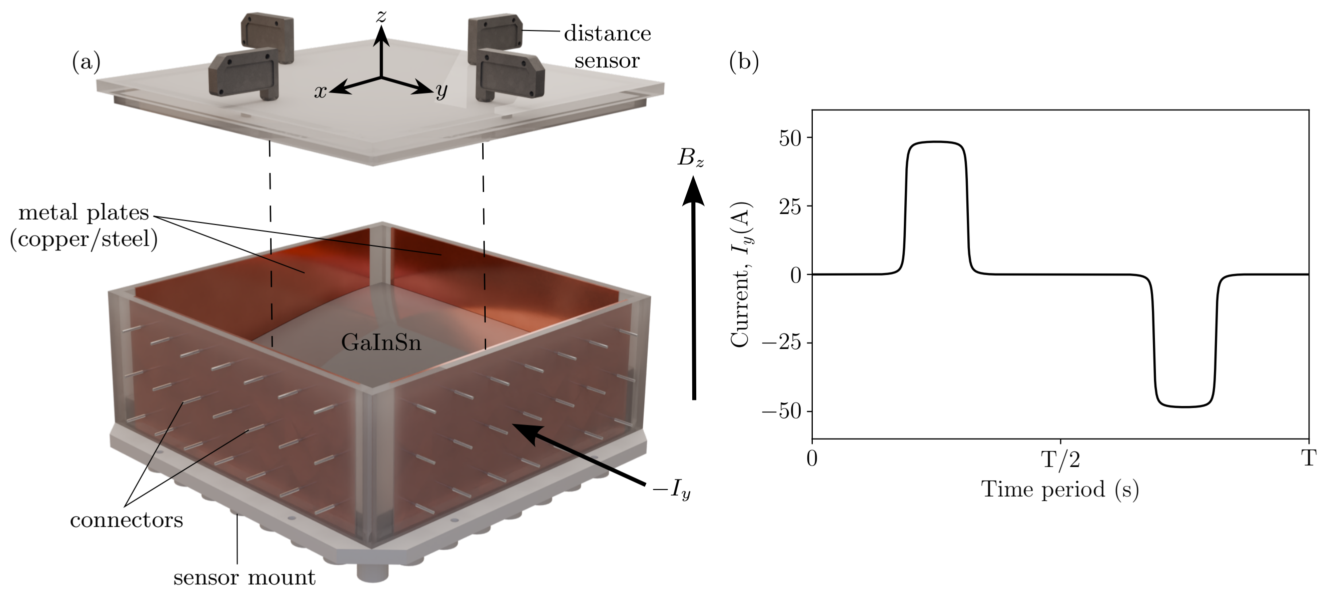


FIG. 9. (a) A 3D rendering of the experimental setup where a standing wave is generated either in the x- or y-direction by passing current I_y or I_x respectively, through the four metal plates on the sidewalls. (b) depicts the current waveform I_y or I_x for a given time period $(2\pi/\omega)$ required to excite a standing wave with resonant frequency ω. The dimensions of the tank are $L_x = 0.255\,\mathrm{m}$, $L_y = 0.255\,\mathrm{m}$ and total height $H = 0.1\,\mathrm{m}$.

inner faces of the four sidewalls had a cut to facilitate the insertion of metal plates $0.23 \times 0.1 \times 0.002\,\mathrm{m}^3$ in size. As shown in Fig. 9(a), the metal plates did not extend fully along the length of the PMMA wall, but had a small gap of $0.0125\,\mathrm{m}$ from either end, to prevent short circuits. The metals in question were copper and stainless steel depending on the required electrical boundary condition, as explained below in Sec. IV B. Each metal plate was welded with 21 uniformly placed shunt resistors, protruding outwards. The combination of thin plates and the shunts ensured a homogeneous current density needed to excite a linear standing wave. The entire tank was vertically placed at the center of a Helmholtz coil generating a quasi-homogeneous vertical magnetic field B_z up to $50\,\mathrm{mT}$ across the wave tank. The tank was filled with GaInSn, a eutectic alloy of Ga, In and Sn which is liquid at room temperature, up to a height of $0.03\,\mathrm{m}$. The rest was filled with Argon gas, to hinder the oxidation of GaInSn due to contact with ambient air. Distance sensors were placed on top of the tank as well as at the bottom, at four locations along the x- and y-axis, to simultaneously measure the instantaneous wave amplitude.

B. The experiment

The free surface of the liquid metal is excited when a horizontal current is supplied to the GaInSn across any two parallel metal plates in the x- or y-axis, in the presence of a constant vertical magnetic field B_z. To generate a quasi-1D standing wave of mode $(1, 0)$ with a resonant frequency ω, a proportional current signal as shown in Fig. 9(b), must be given in the y-direction. The time period of the signal $T = 2\pi/\omega$ and the size of the wave is proportional to the amplitude of the signal. Because we magnetohydrodynamically excite the wave, we are limited to using electrically conducting materials in the sidewalls. To simulate conducting and insulating sidewalls, we make use of copper (Cu-ETP) and nickel-plated stainless steel. As given in Table IV, GaInSn is significantly less conductive than copper, by a factor of about twenty, and has a conductivity that is three times that of stainless steel. Therefore, copper walls act as a relative conductor (B.C 1) and stainless steel walls as a relative insulator (B.C 4).

We conduct the experimental measurements between Ha = 100–500 (B_z = 10–50 mT). From Fig. 6, we know that at lower values of Ha, the effect of the magnetic field on the velocity gradient at the wall boundary layer is negligible. This means that the dissipation due to boundary effects can be considered as a purely hydrodynamic phenomenon and that any field-induced change is dominated by Ohmic dissipation. This allows us to isolate the contribution of Ohmic dissipation to the total damping measured, which is one of the main results of the present study. The following steps were used to achieve this: i) we first excited a wave using an oscillating current signal I_y and magnetic field B_z. After a certain amount of time ($t = 60\,\mathrm{s}$) after the wave was properly developed, both the current and magnetic

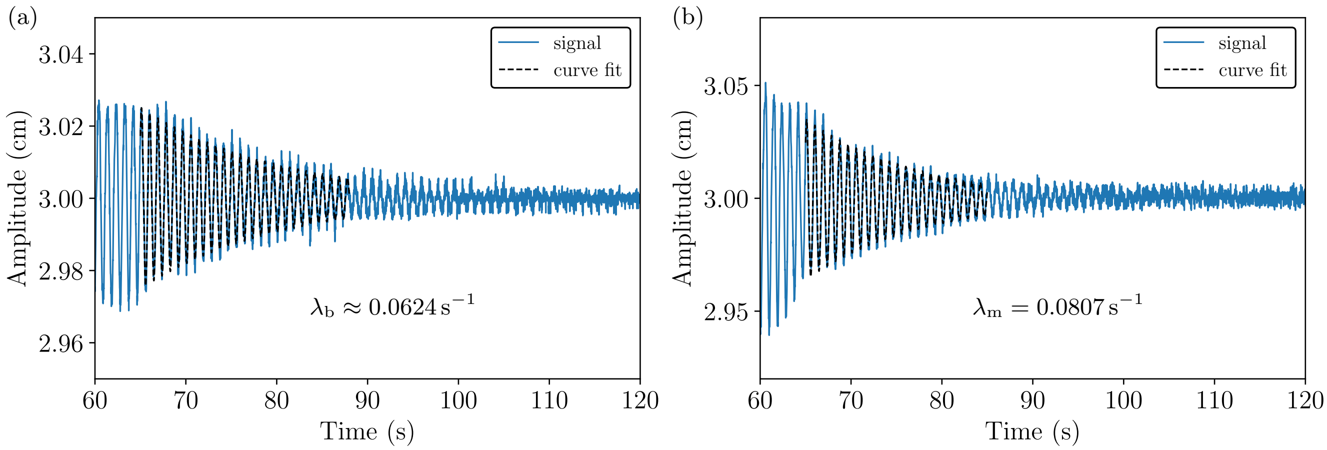


FIG. 10. The amplitude decay profile of wave mode $(1, 0)$ in a square wave tank with stainless steel sidewall inserts, a) without the presence of a vertical magnetic field B_z and horizontal current I_y and b) in the presence of a magnetic field $B_z = 20\,\mathrm{mT}$ (Ha = 196) without horizontal current. At low Ha, we can ignore the effects of the Hartmann layer and consider the damping contribution at the boundary to be purely hydrodynamic. The difference between the total damping and the boundary layer contributions yields the damping rate due to Ohmic dissipation $\lambda_o = 0.018\,s^{-1}$.

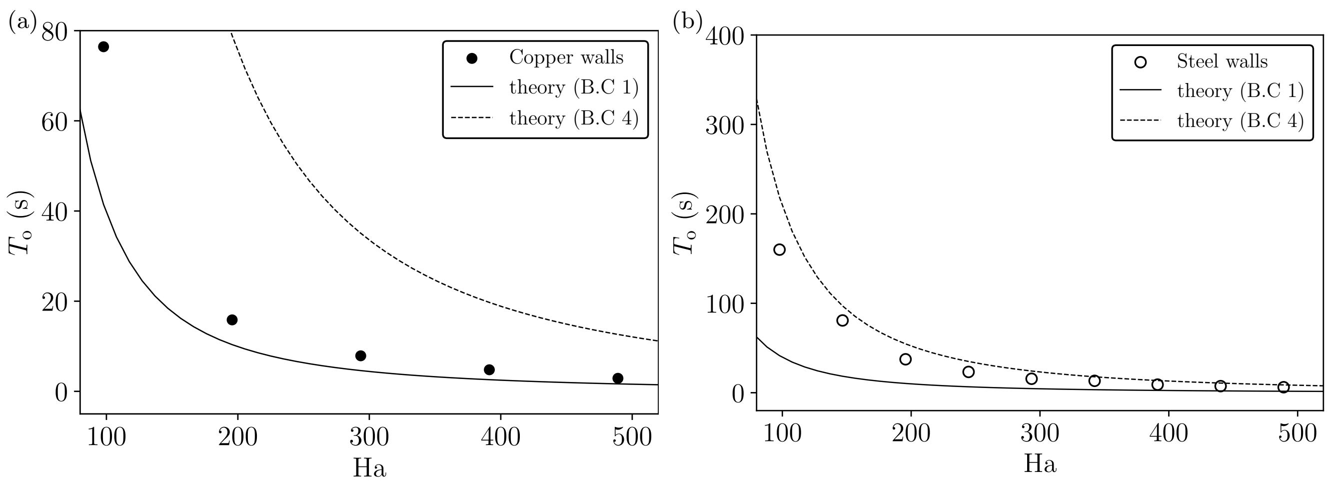


FIG. 11. The comparison of theoretical and experimental damping times $T_o = \lambda_o^{-1}$ in a wave tank with a) copper sidewalls (B.C 1), and b) stainless steel sidewalls (B.C 4) at different Ha. For the theoretical calculation, copper walls are treated as perfect conductors and stainless steel walls as perfect insulators.

fields were shut off and the amplitude of the wave was recorded with respect to time. The slope of this decaying signal gives us the viscous damping rate λ_v, as the energy of the wave is dissipated solely due to viscous effects at the boundary. ii) Subsequently, a wave was excited again, but only the input current I_y was cut off after a given time to initiate wave decay. The slope of the decaying amplitude signal gives the magnetic damping rate λ_m, which takes into account both the Ohmic dissipation and the boundary layer effects [Eq. (49)]. We have already established that the boundary layer effects with and without the magnetic field are the same, $\lambda_b \approx \lambda_v$. The difference $\lambda_m - \lambda_v$ is the Ohmic damping rate λ_o and the inverse $T_o = \lambda_o^{-1}$, the Ohmic damping time for a particular Ha. Fig. 10(a) and 10(b) provides an example of the extraction of λ_o from λ_b and λ_m respectively at Ha = 196 ($B_z = 20\,\mathrm{mT}$) in a tank with stainless steel sidewalls. The Ohmic damping rate/time is independent of the initial amplitude of the wave.

Fig. 11 contains the experimental results and compares T_o measurements in a tank with copper and stainless steel sidewalls, at different values of Ha with the theoretical cases B.C 1 and B.C 4. We present the results in terms of damping times in order to remain consistent with the previous literature [14]. We saw good agreement in both cases, where the measurements done for the waves in copper and stainless steel sidewalls closely follow the theoretical damping times of B.C 1 and B.C 4 cases respectively. As seen in Fig. 11(a) and Fig. 11(b), the slight overshoot in the measurements with copper walls and the undershoot with stainless steel walls was expected. The metal inserts have a certain amount of thickness ($d = 2\,\mathrm{mm}$) and are neither perfect conductors nor insulators.

The damping times at higher depths, i.e. $h = 0.04, 0.05$ and $0.06\,\mathrm{m}$ were also measured, but revealed no considerable change in the values. It was not possible to achieve greater depths due to limited wall height of the tank, and the risk of GaInSn splashing outward and damaging the top-mounted distance sensors.

V. CONCLUDING REMARKS

In the present study, we formulate a theory on magnetic damping of liquid metals in non-shallow systems, which has grave significance in the electromagnetic processing of materials. The damping behavior of magnetic fields has been observed in certain industrial devices, such as induction heaters and in specific processes like continuous casting of steel and aluminum. However, this has not yet been fully realized theoretically. Prior studies have taken shallow water approximation and formulated damping theory for simpler wave modes in a system with simpler electrical boundary conditions; creating some gaps in the theory which our work has sought to address.

To calculate the magnetic damping rates, we have taken a electrically conducting liquid system with four different electrical boundary conditions, ranging from fully conducting walls, to fully insulating walls along with mixed conditions in between. The magnetic damping rate can be classified to have bulk and boundary contributions. The former is due to Ohmic dissipation. To calculate the Ohmic damping rate, we first derived an exact solution for the electric potential of the induced eddy currents and compared the eddy current behavior in all the four cases. The currents close in the perfectly conducting walls and close within the liquid bath near perfect insulators. For an arbitrary bidirectional standing wave of mode (m, n), the number of 2D eddies induced were found to be exactly equal to $(m + 1)(n + 1)$. The damping formulation consists of a pure induction term $\mathcal{Q}_{\mathrm{ind}}$ which is the Ohmic dissipation in the liquid bath when all surrounding walls are perfect conductors and a very intricate correction term $\mathcal{Q}_{\mathrm{corr}}$ which originates due to the insulating walls. The difference between the two terms is the net Ohmic dissipation in a system and the ratio between the difference and the energy of the wave is the Ohmic damping rate λ_{o}. The electrical boundary condition influences the path taken by the eddy currents within the liquids which in turn influences the overall dissipation of energy in the system. Secondly, although smaller in magnitude, the dissipation due to boundary effects is due to the transition of the oscillating Stokes boundary layer at the walls into an electromagnetic boundary layer. The boundaries perpendicular to the orientation of the magnetic field see the presence of the Hartmann layer, where those parallel, see the Shercliff layer. We have formulated the thickness of the Hartmann and the Shercliff boundary layer as well as the damping rate λ_{b}. As outlined in the supplementary material, all derivations of the formulae are comprehensively detailed, along with a Python module that incorporates the complete set of equations.

Our formulations found good agreement with the theoretical and experimental results of Sreenivasan *et al.* [14], for the damping of a shallow, quasi-1D wave of mode $(1, 0)$ in an electrically insulating rectangular tank. We also validated our model at a non-shallow depth using the experimental setup of Hegde *et al.* [18], where we recorded the exponential decay of a magnetohydrodynamically excited free surface wave. Unlike [14], we operated at lower values of Ha, which allowed us to isolate the damping contributions solely due to Ohmic dissipation.

Although, the analytical formulation of the damping rates/damping times is presented in a single-layer form, it can also be applicable to two-layer systems with large conductivity differences. Here, only the liquid with the larger electrical conductivity contributes to Ohmic dissipation. The Hartmann layer also forms at the interface (on the side of the better electric conductor), where the viscous stresses balance the Lorentz stresses. This is modeled in the Appendix . Together, this can further serve as an easy benchmark for direct numerical solvers. For example, it can be used to iteratively obtain the electric potential V from the Poisson equation $\nabla^2 V = \boldsymbol{\nabla} \cdot (\boldsymbol{U} \times \boldsymbol{B})$, which is often challenging.

However, certain questions have yet to be resolved. First, the experimental setup is incapable of exciting waves with higher wave modes, a key feature of our analytical model. The damping behavior of bidirectional wave modes $(m > 0, n > 0)$ is dictated by complex closing current patterns and therefore much more complicated, making it very rewarding to set up dedicated liquid metal wave damping experiments. Second, the effect of mixed electrical boundary conditions on the damping rate have not be validated due to lack of existing studies and the intricacies involved in constructing an experimental setup with such conditions. Conclusive clarification of this requires the construction of a new experiment along with the use of computational fluid dynamics.

ACKNOWLEDGMENTS

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Appendix: Modeling the Hartmann layer at the interface of a two-layer stratified fluid system

In a two-layer stratified system with varying density $\rho_1 < \rho_2$ and large conductivity difference $\sigma_1 \ll \sigma_2$, the phenomenon of induction largely remains the same as outlined in Secs. II B and II C. However, the modeling of boundary layer dissipation is distinct due to the incorporation of an additional boundary, the interface. A free surface is allowed to slide freely and therefore, has zero shear stresses. This is not the case in an interface between two liquid mediums. Here, tangential velocity and tangential stress are continuous, but the velocity gradients generally differ because the dynamic viscosities differ. This leads to the formation of a interfacial Hartmann layer where the tangential induced currents coupled with the vertical magnetic field $B_z \boldsymbol{e}_z$ produce Lorentz forces which balance the mechanical tangential stresses. In liquid-liquid systems where $\sigma_1 \ll \sigma_2$, the Hartmann layer forms at the interface of the better electrical conductor. To model the dissipation of energy due to the interfacial Hartmann layer, we write the following balance equations:

$$\boldsymbol{u}_{1,\perp}|_{\mathcal{S}} = \boldsymbol{u}_{2,\perp}|_{\mathcal{S}}, \tag{A.1a}$$

$$p_2|_{\mathcal{S}} - p_1|_{\mathcal{S}} = (\rho_2 - \rho_1)g\eta + 2(\rho_2\nu_2\partial_z u_{2,z} - \rho_1\nu_1\partial_z u_{1,z}), \tag{A.1b}$$

$$\rho_1\nu_1(\partial_z u_{1,x} + \partial_x u_{1,z})|_{\mathcal{S}} = \rho_2\nu_2(\partial_z u_{2,x} + \partial_x u_{2,z})|_{\mathcal{S}}, \tag{A.1c}$$

$$\rho_1\nu_1(\partial_z u_{1,y} + \partial_y u_{1,z})|_{\mathcal{S}} = \rho_2\nu_2(\partial_z u_{2,y} + \partial_y u_{2,z})|_{\mathcal{S}}. \tag{A.1d}$$

Here, Eq. (A.1a) refers to the continuity of the tangential velocity at the interface $\mathcal{S}$, where the suffix ($\perp$) denotes the tangential component of each fluid layer. Secondly, Eq. (A.1b) is the modified Young-Laplace equation which captures the hydrostatic balance considering normal stresses at the interface. Lastly, Eqs. (A.1c–A.1d) show the continuity of the tangential stresses at the interface.

To deduce the tangential velocities in the Hartmann layer on either sides of the interface, we make use of the boundary conditions (A.1c and A.1d) to satisfy

$$\boldsymbol{u}_{1,\perp} + \bar{\boldsymbol{u}}_{1,\perp}|_{\mathcal{S}} \approx \boldsymbol{u}_{2,\perp} + \bar{\boldsymbol{u}}_{2,\perp}|_{\mathcal{S}} \qquad \begin{cases} \rho_1\nu_1\partial_z \bar{u}_{1,x}|_{\mathcal{S}} \approx \rho_2\nu_2\partial_z \bar{u}_{2,x}|_{\mathcal{S}} \\ \rho_1\nu_1\partial_z \bar{u}_{1,y}|_{\mathcal{S}} \approx \rho_2\nu_2\partial_z \bar{u}_{2,y}|_{\mathcal{S}} \end{cases}, \tag{A.2}$$

and subsequently obtain

$$\bar{\boldsymbol{u}}_{1,\perp}|_{\mathcal{S}} = \frac{\omega\Lambda}{k\sqrt{\rho_1\nu_1\sigma_1}\sqrt{1+\sqrt{1+\tau_1^2\omega^2}}} \begin{Bmatrix} \dfrac{m\pi}{L_x}\sin\left[\dfrac{m\pi}{L_x}\left(x+\dfrac{L_x}{2}\right)\right]\cos\left[\frac{n\pi}{L_y}\left(y+\dfrac{L_y}{2}\right)\right]\boldsymbol{e}_x \\ \dfrac{n\pi}{L_y}\cos\left[\dfrac{m\pi}{L_x}\left(x+\dfrac{L_x}{2}\right)\right]\sin\left[\dfrac{n\pi}{L_y}\left(y+\dfrac{L_y}{2}\right)\right]\boldsymbol{e}_y \end{Bmatrix} \mathrm{e}^{-\frac{1}{\delta_{\mathrm{tot}}}z}, \tag{A.3}$$

$$\bar{\boldsymbol{u}}_{2,\perp}|_{\mathcal{S}} = \frac{-\omega\Lambda}{k\sqrt{\rho_2\nu_2\sigma_2}\sqrt{1+\sqrt{1+\tau_2^2\omega^2}}} \begin{Bmatrix} \dfrac{m\pi}{L_x}\sin\left[\dfrac{m\pi}{L_x}\left(x+\dfrac{L_x}{2}\right)\right]\cos\left[\frac{n\pi}{L_y}\left(y+\dfrac{L_y}{2}\right)\right]\boldsymbol{e}_x \\ \dfrac{n\pi}{L_y}\cos\left[\dfrac{m\pi}{L_x}\left(x+\dfrac{L_x}{2}\right)\right]\sin\left[\dfrac{n\pi}{L_y}\left(y+\dfrac{L_y}{2}\right)\right]\boldsymbol{e}_y \end{Bmatrix} \mathrm{e}^{\frac{1}{\delta_{\mathrm{tot}}}z}, \tag{A.4}$$

where,

$$\delta_{\mathrm{tot}} = \frac{\sqrt{2\tau_i\nu_i}}{\sqrt{1+\sqrt{1+\tau_i^2\omega^2}}}, \qquad \text{(from Eq. (40))}, \tag{A.5}$$

and Λ given by

$$\Lambda = A_{mn}\,\frac{(\coth[kh_1] + \coth[kh_2])}{\dfrac{1}{\sqrt{\rho_1\nu_1\sigma_1}\sqrt{1+\sqrt{1+\tau_1^2\omega^2}}} + \dfrac{1}{\sqrt{\rho_2\nu_2\sigma_2}\sqrt{1+\sqrt{1+\tau_2^2\omega^2}}}}. \tag{A.6}$$

The tangential velocities help us in determining the tangential stresses which can be related to the strain rate. We use the same approach of Hegde *et al.* [21] for a two-layer stratified system undergoing linear sloshing. Therefore, the new mechanical energy balance reads:

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(\sum_{i=1,2}\frac{1}{2}\int_{V_i}\rho_i\|\boldsymbol{u}_i\|^2\mathrm{d}V\right)+\frac{\mathrm{d}}{\mathrm{d}t}\left(\int_{\mathcal{S}}\frac{1}{2}(\rho_2-\rho_1)g\eta^2\,\mathrm{d}\mathcal{S}\right)=-\sum_{i=1,2}2\rho_i\nu_i\int_{V_i}\boldsymbol{\varepsilon}_i:\boldsymbol{\varepsilon}_i\,\mathrm{d}V, \tag{A.7}$$

which is same as Eq. (45) but with the addition of the potential energy term arising due to the interface. In a standing wave, the kinetic and potential energy have a phase difference of $\pi/2$. Solving the right hand side of Eq. (A.7) as shown in Sec. III of the supplementary material [22], we get the damping rate at the interface:

$$\lambda_{\mathrm{int}}=\frac{k}{2\sqrt{2}}\left(\frac{\dfrac{1}{\xi_1\sigma_1\sqrt{\tau_1\nu_1}}+\dfrac{1}{\xi_2\sigma_2\sqrt{\tau_2\nu_2}}}{\dfrac{\rho_1}{\tanh[kh_1]}+\dfrac{\rho_2}{\tanh[kh_2]}}\right)\left(\frac{\dfrac{1}{\tanh[kh_1]}+\dfrac{1}{\tanh[kh_2]}}{\dfrac{1}{\sqrt{\rho_1\nu_1\sigma_1}\,\xi_1}+\dfrac{1}{\sqrt{\rho_2\nu_2\sigma_2}\,\xi_2}}\right)^2, \tag{A.8}$$

where,

$$\xi_i=\sqrt{1+\sqrt{1+\tau_i^2\omega^2}},\quad \text{and}\quad \tau_i=\frac{\rho_i}{\sigma_i B_z^2},\quad \{i=1,2\}. \tag{A.9}$$

Eq. (A.8) is the damping rate due to viscous and magnetohydrodynamic boundary layers at the interface of two-liquids with large conductivity difference. The upper layer of the interface is purely viscous whereas the lower layer constitutes the viscous and Hartmann layer.

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